

BFS Traversal

Step 1: Start at E

Legend

Complete

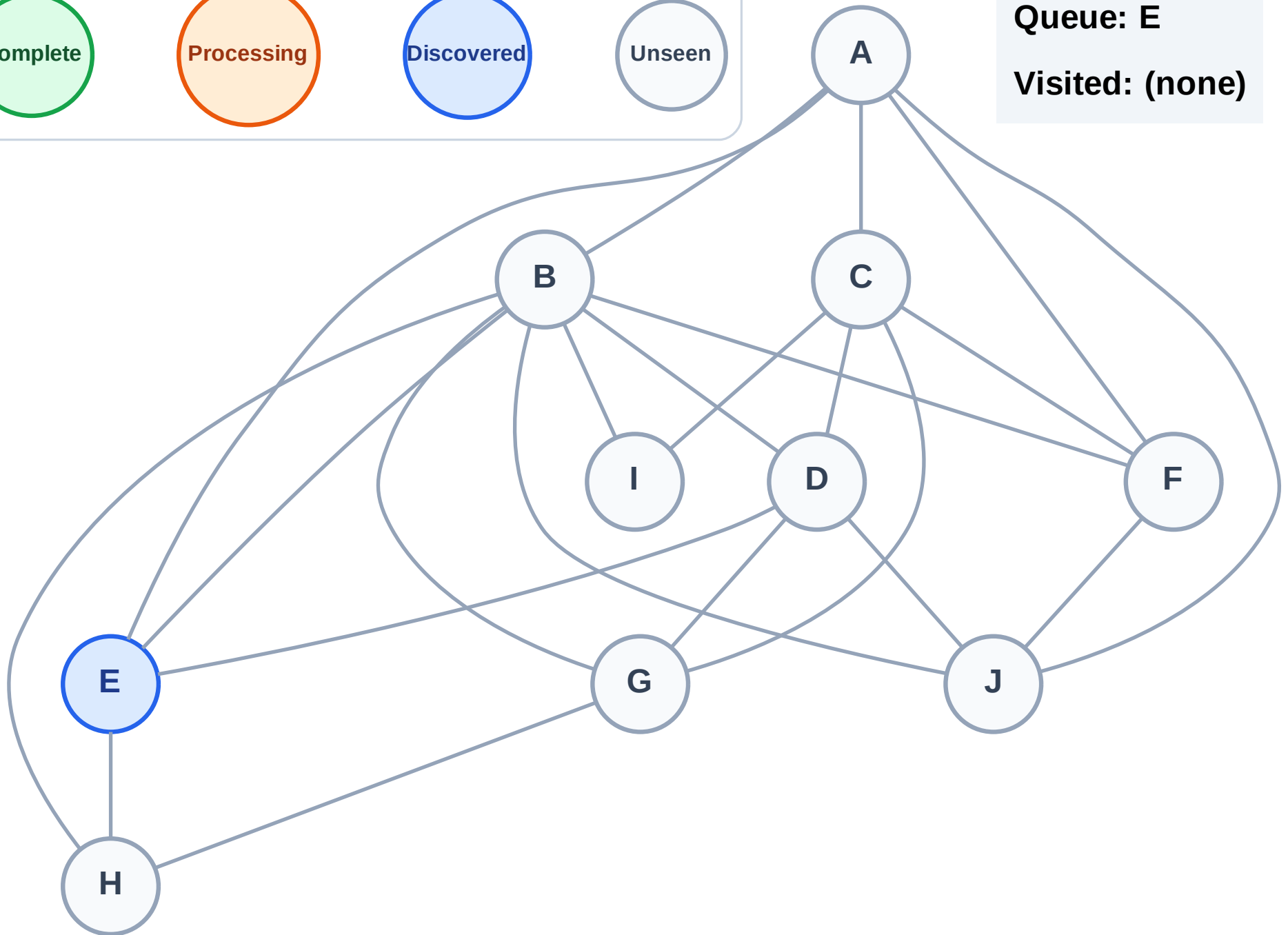
Processing

Discovered

Unseen

Queue: E

Visited: (none)



BFS Traversal

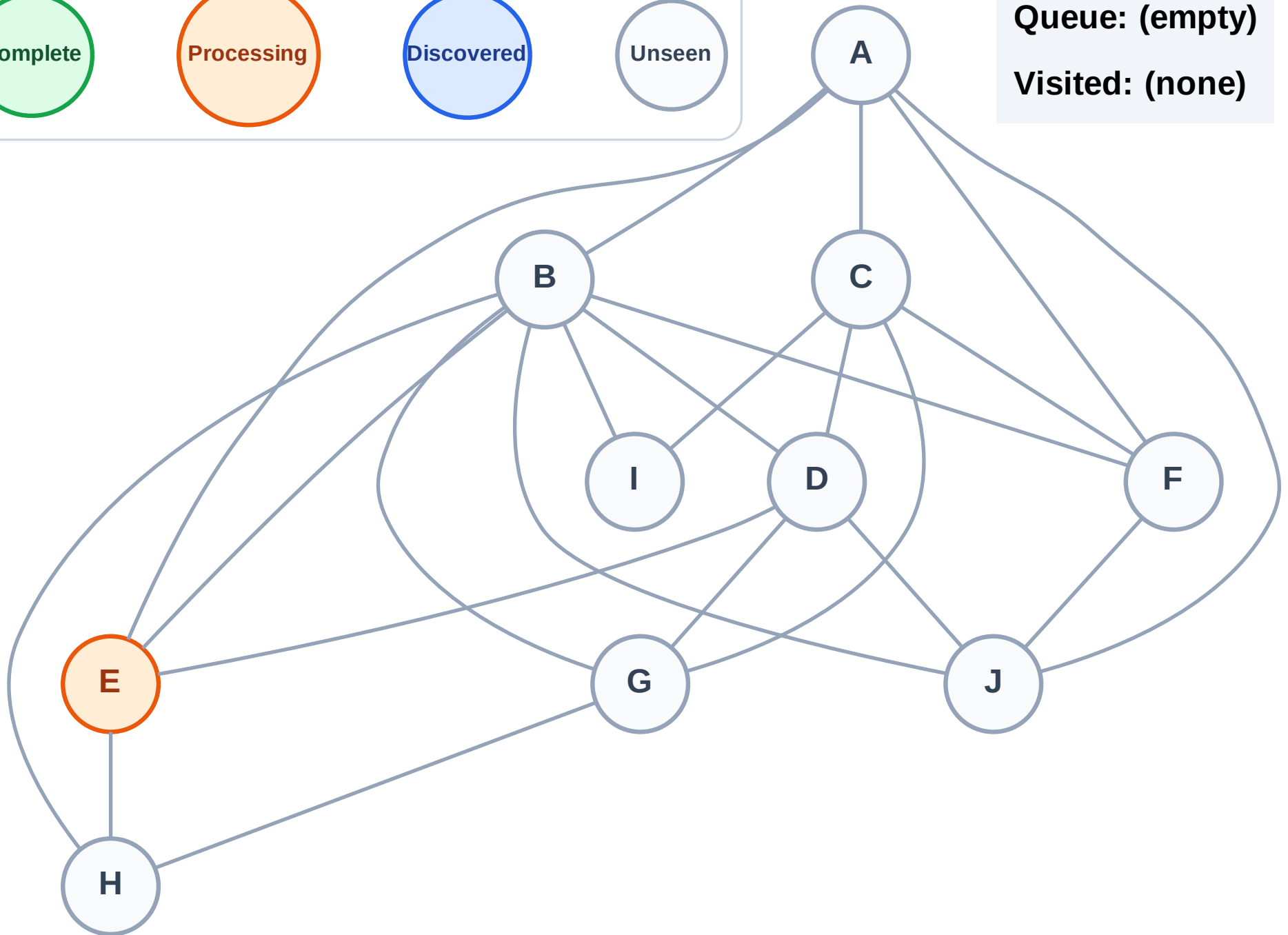
Step 2: Process node E

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

Step 3: Discover A, B, D, H from E

Legend

Complete

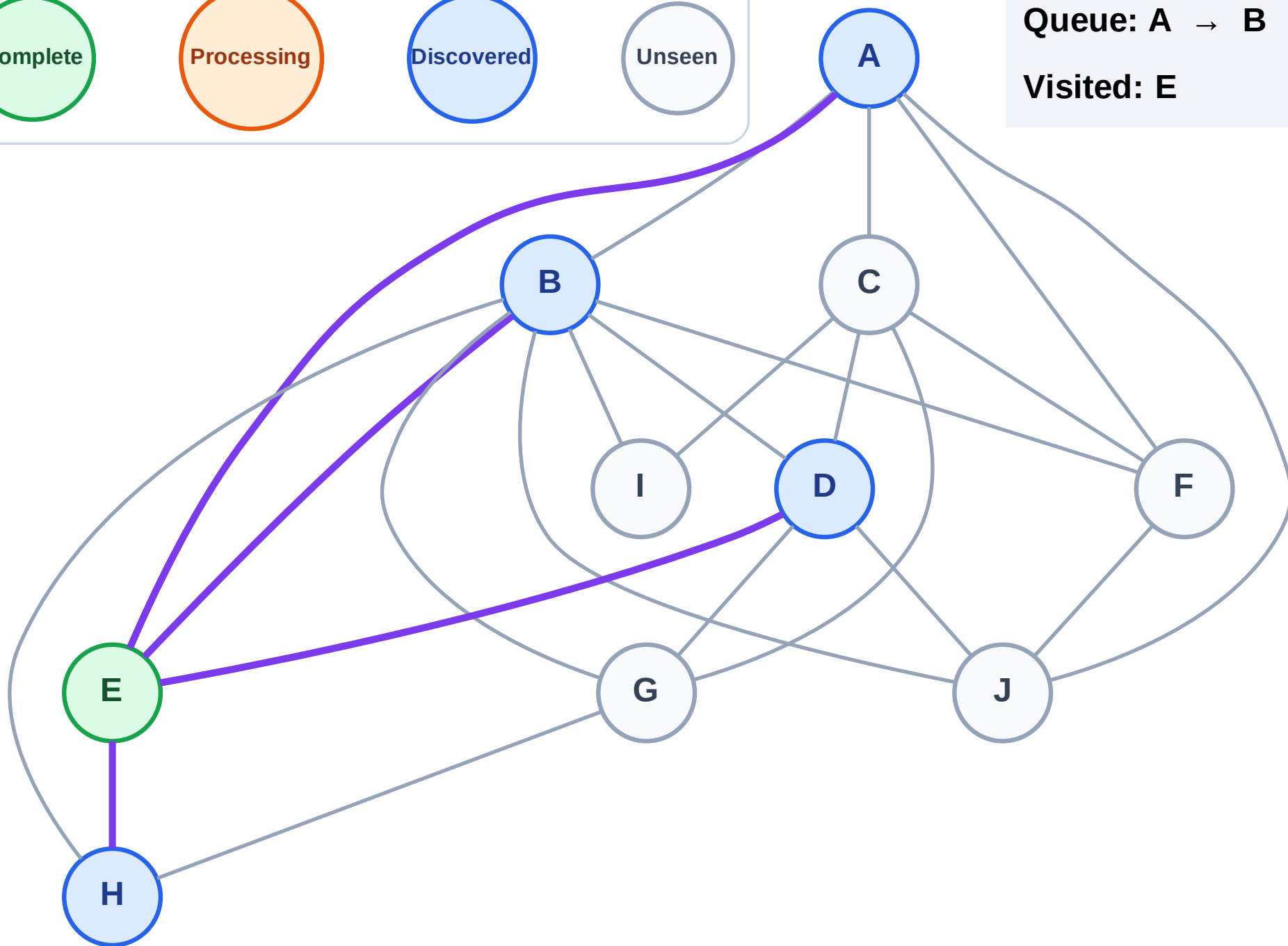
Processing

Discovered

Unseen

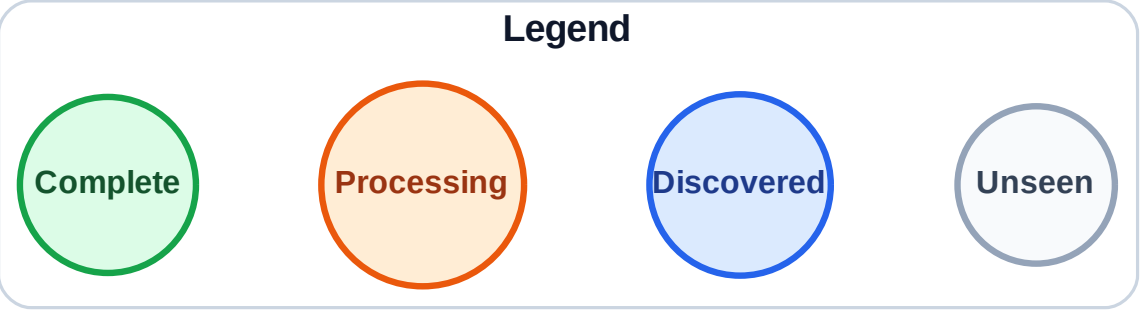
Queue: A → B → D → H

Visited: E

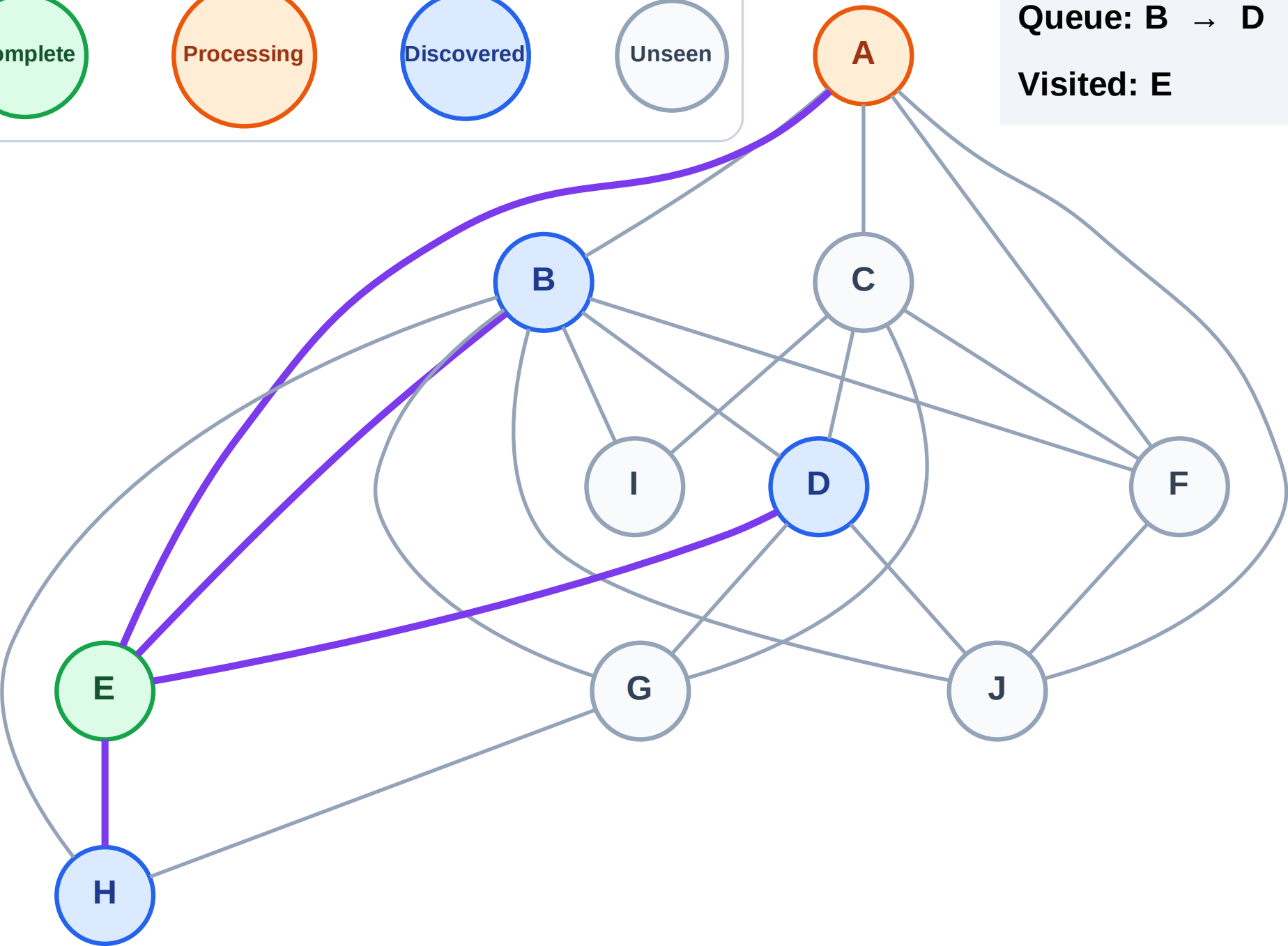


BFS Traversal

Step 4: Process node A



Queue: B → D → H
Visited: E



BFS Traversal

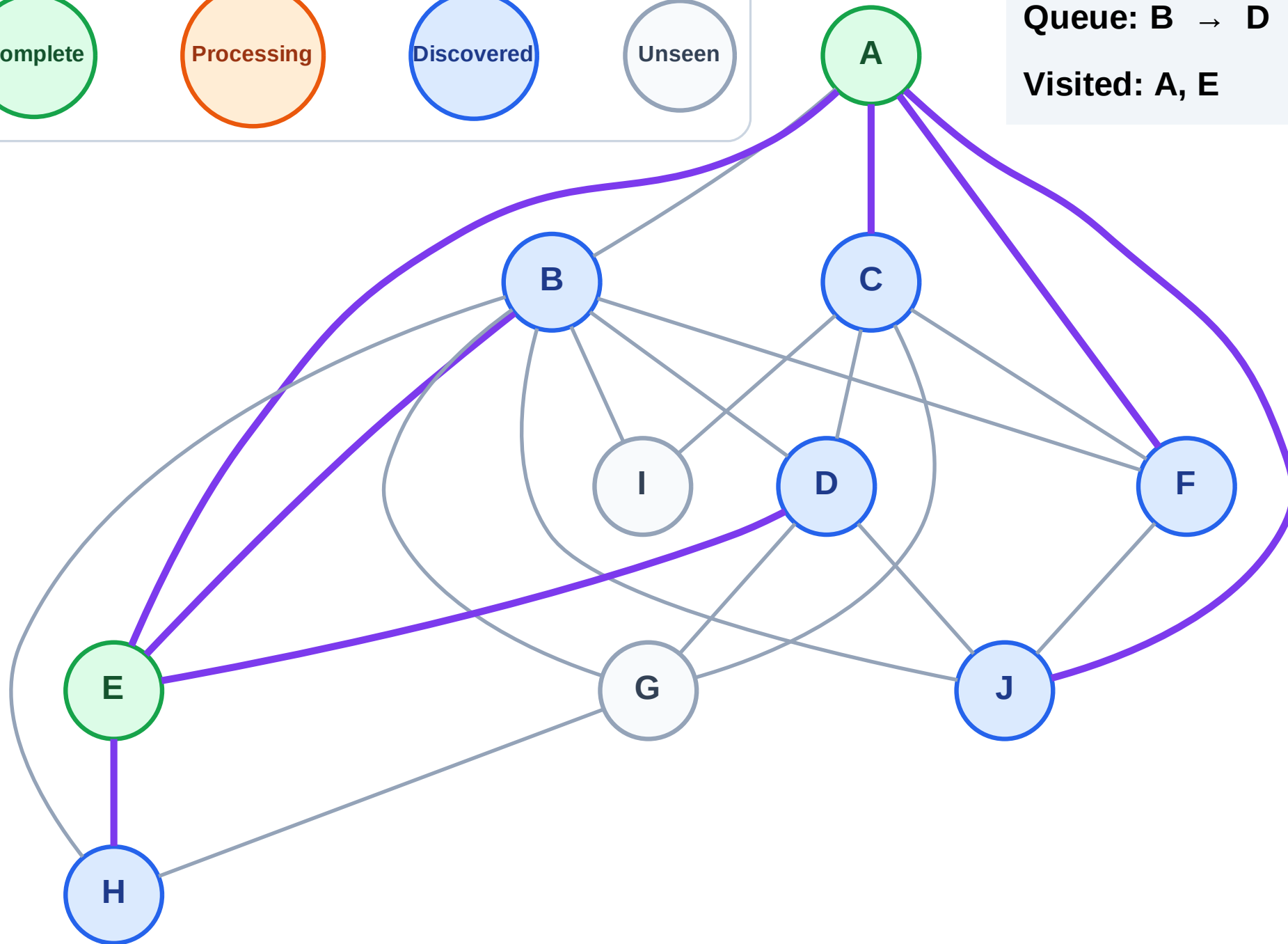
Step 5: Discover C, F, J from A

Legend



Queue: B → D → H → C → F → J

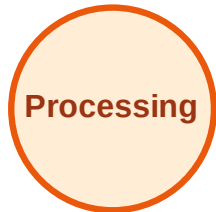
Visited: A, E



BFS Traversal

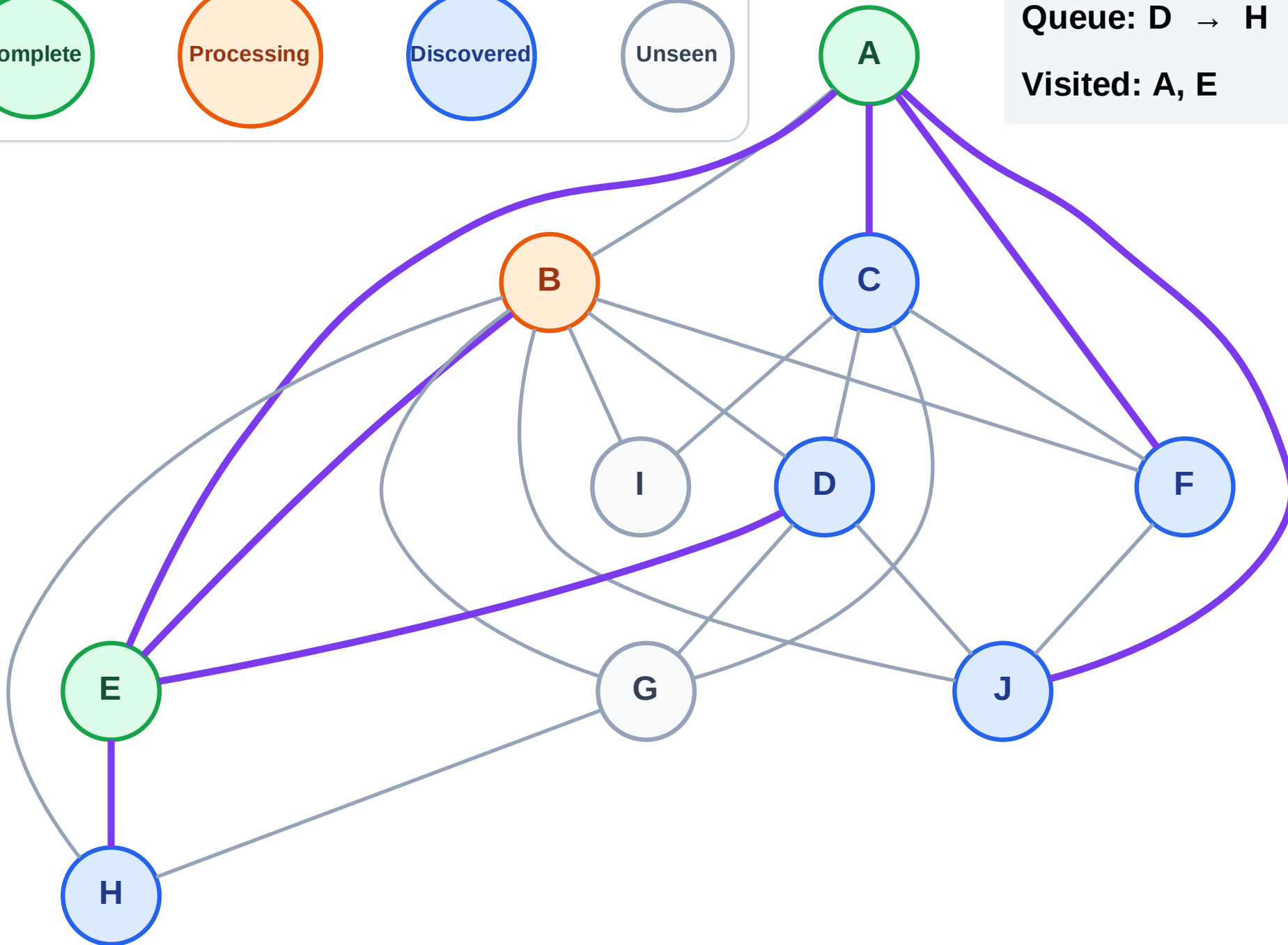
Step 6: Process node B

Legend



Queue: D → H → C → F → J

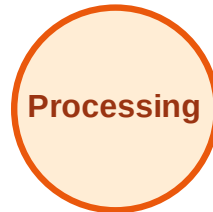
Visited: A, E



BFS Traversal

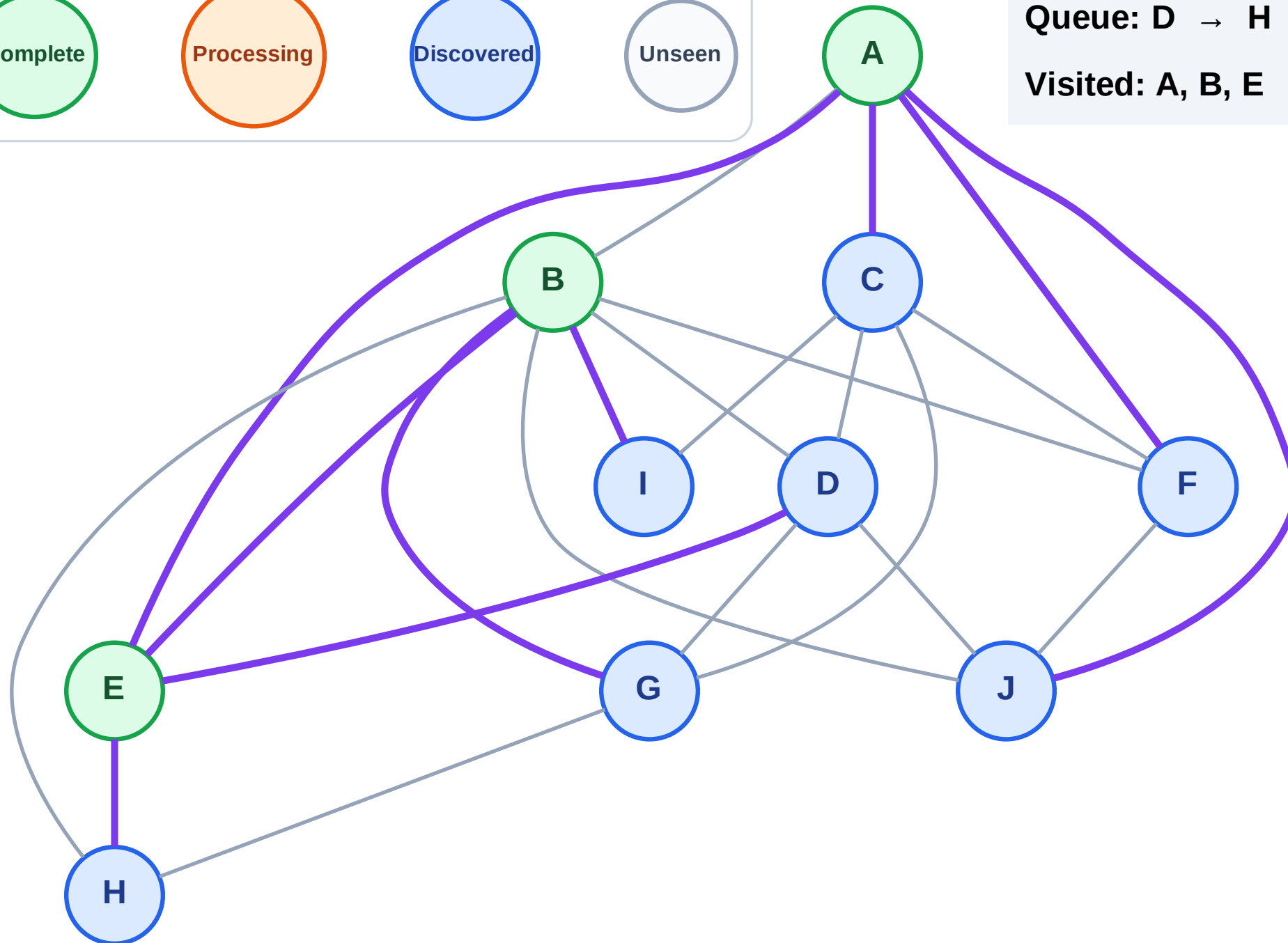
Step 7: Discover G, I from B

Legend



Queue: D → H → C → F → J → G → I

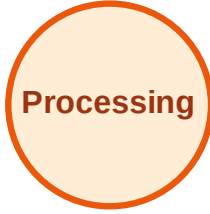
Visited: A, B, E



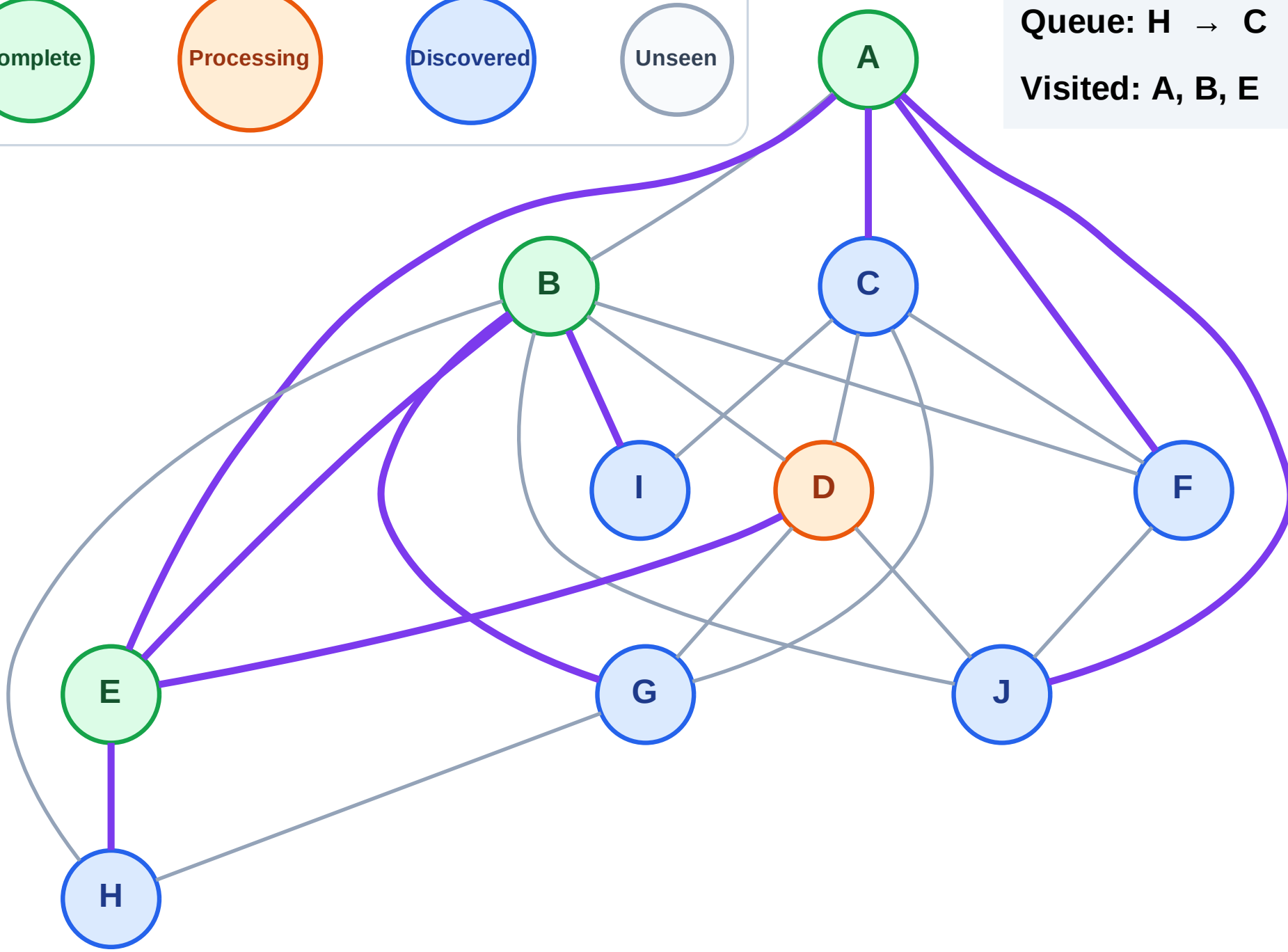
BFS Traversal

Step 8: Process node D

Legend



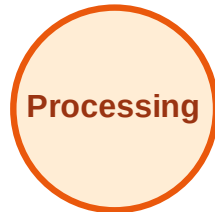
Queue: H → C → F → J → G → I
Visited: A, B, E



BFS Traversal

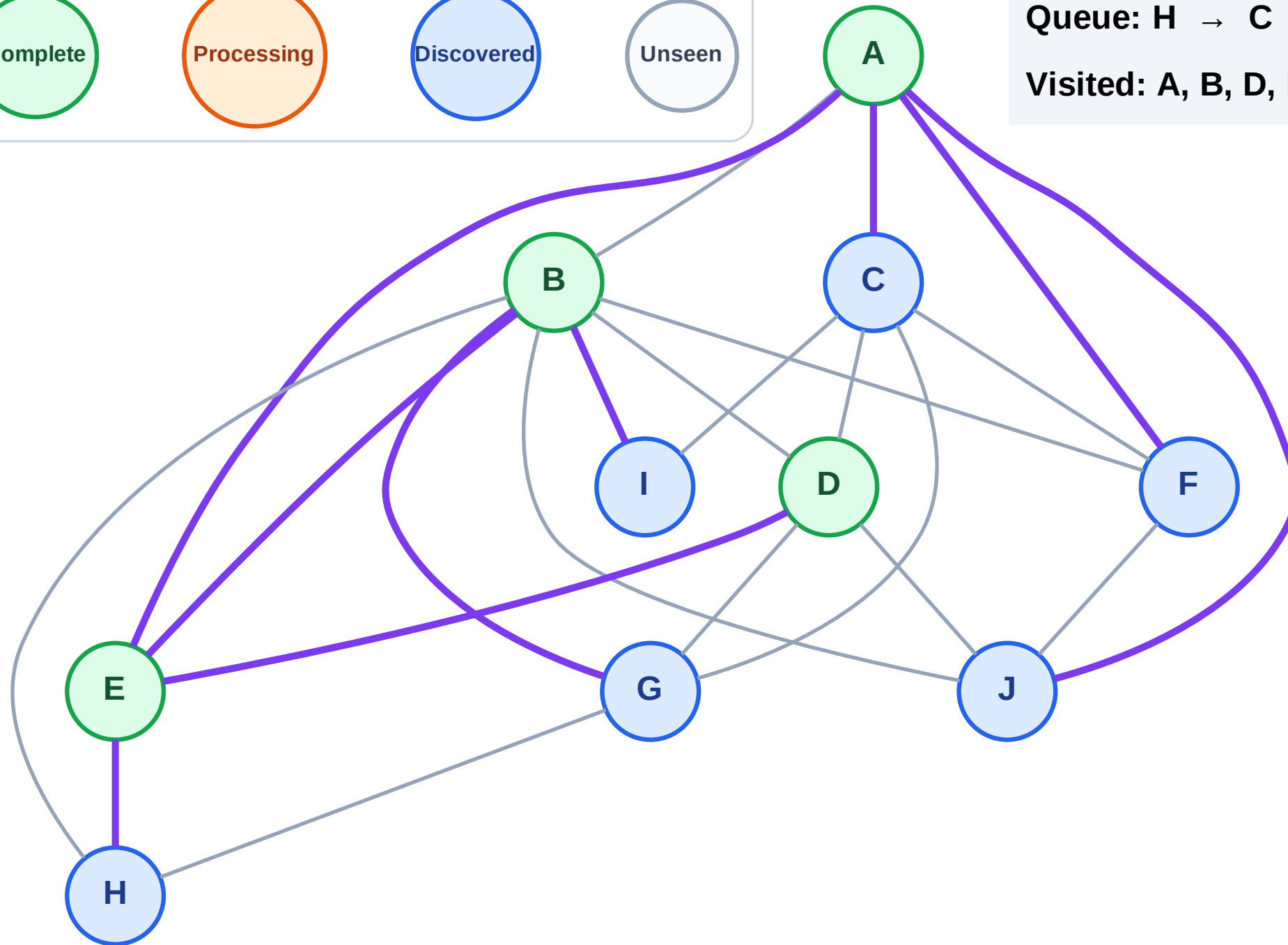
Step 9: No new neighbors from D

Legend



Queue: H → C → F → J → G → I

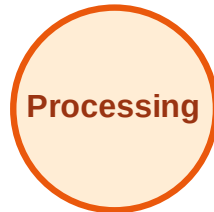
Visited: A, B, D, E



BFS Traversal

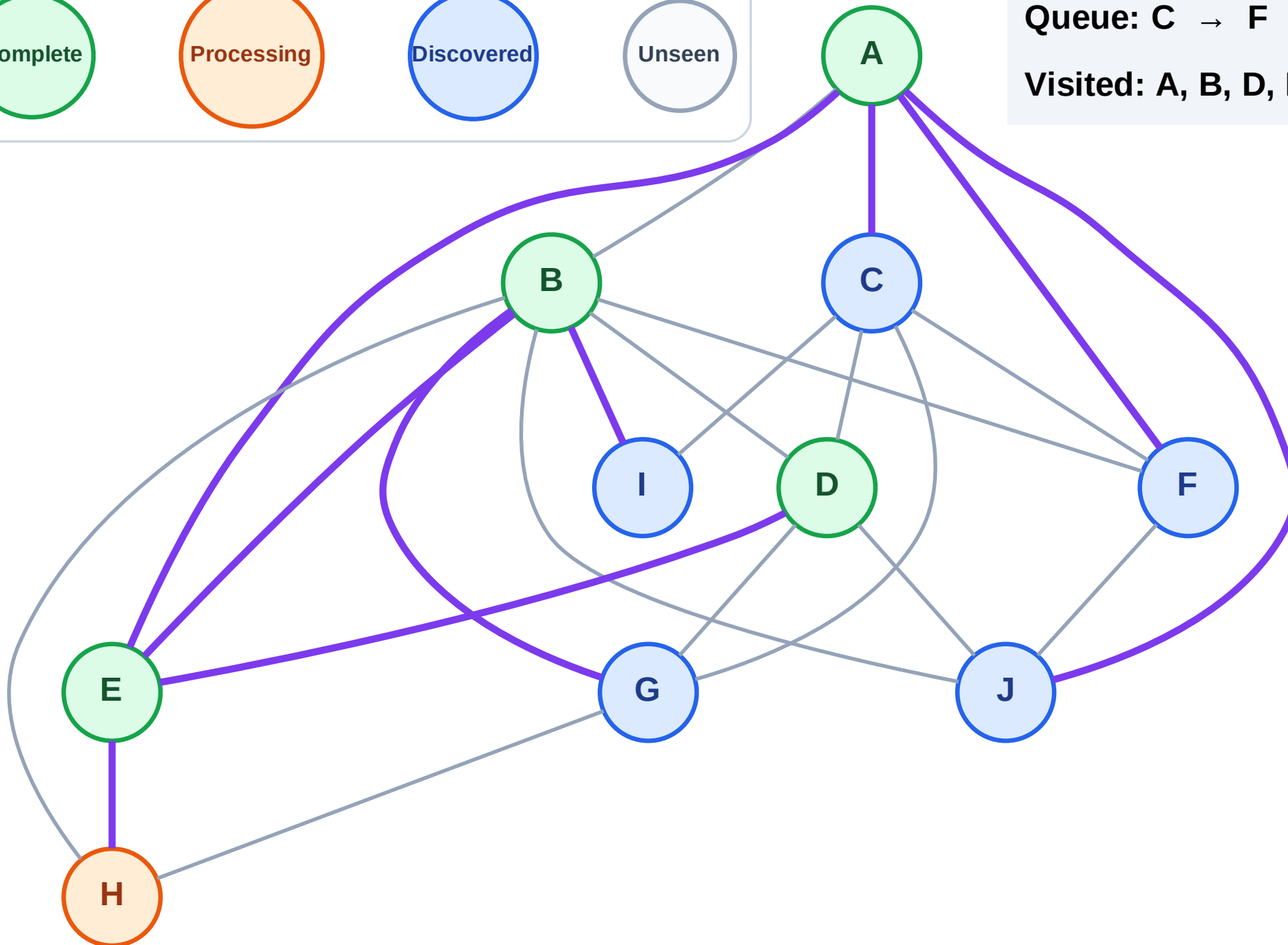
Step 10: Process node H

Legend



Queue: C → F → J → G → I

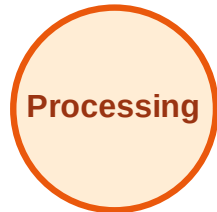
Visited: A, B, D, E



BFS Traversal

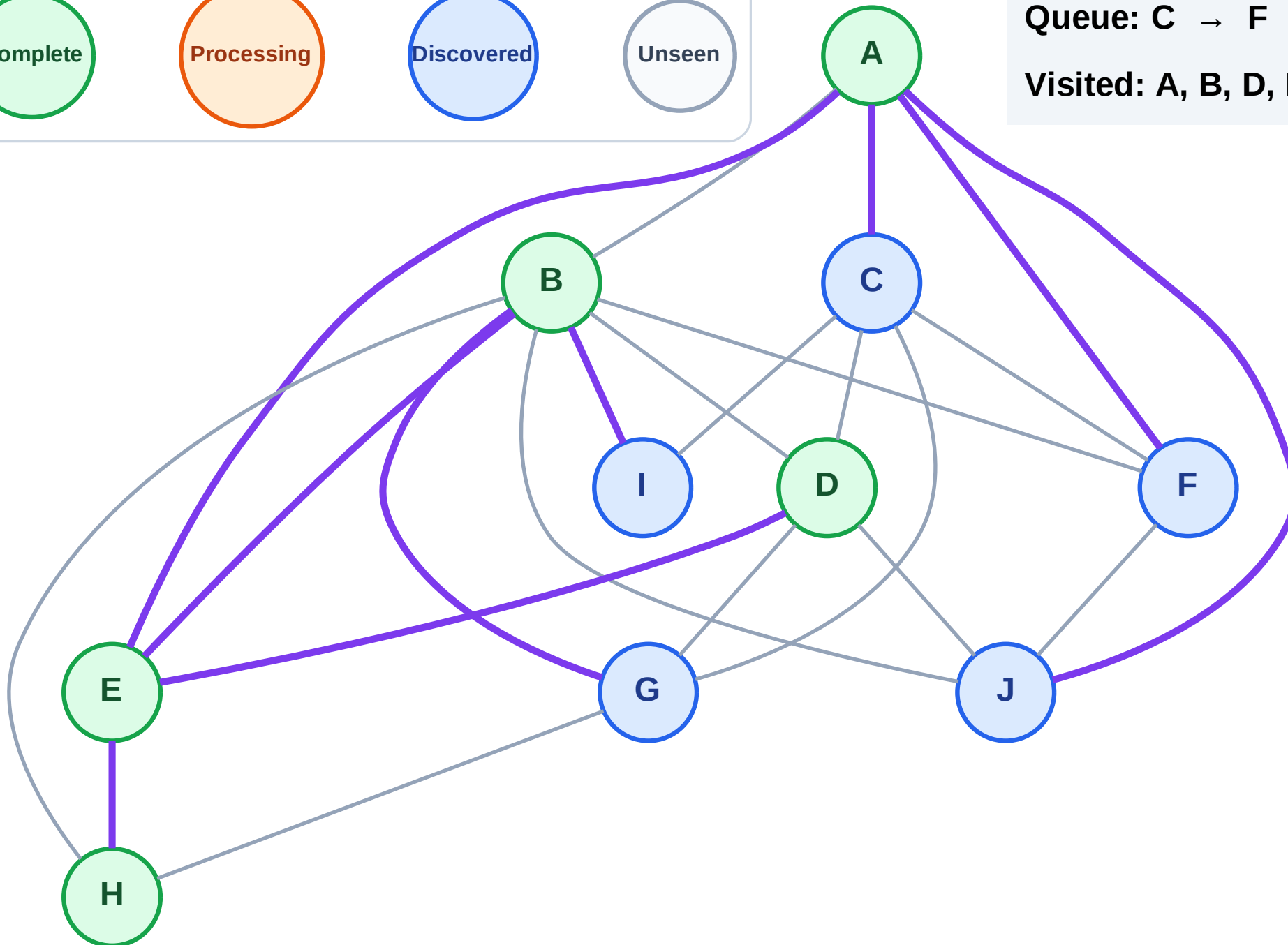
Step 11: No new neighbors from H

Legend



Queue: C → F → J → G → I

Visited: A, B, D, E, H



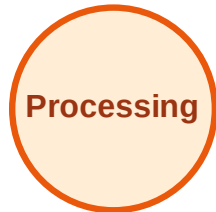
BFS Traversal

Step 12: Process node C

Legend



Complete



Processing



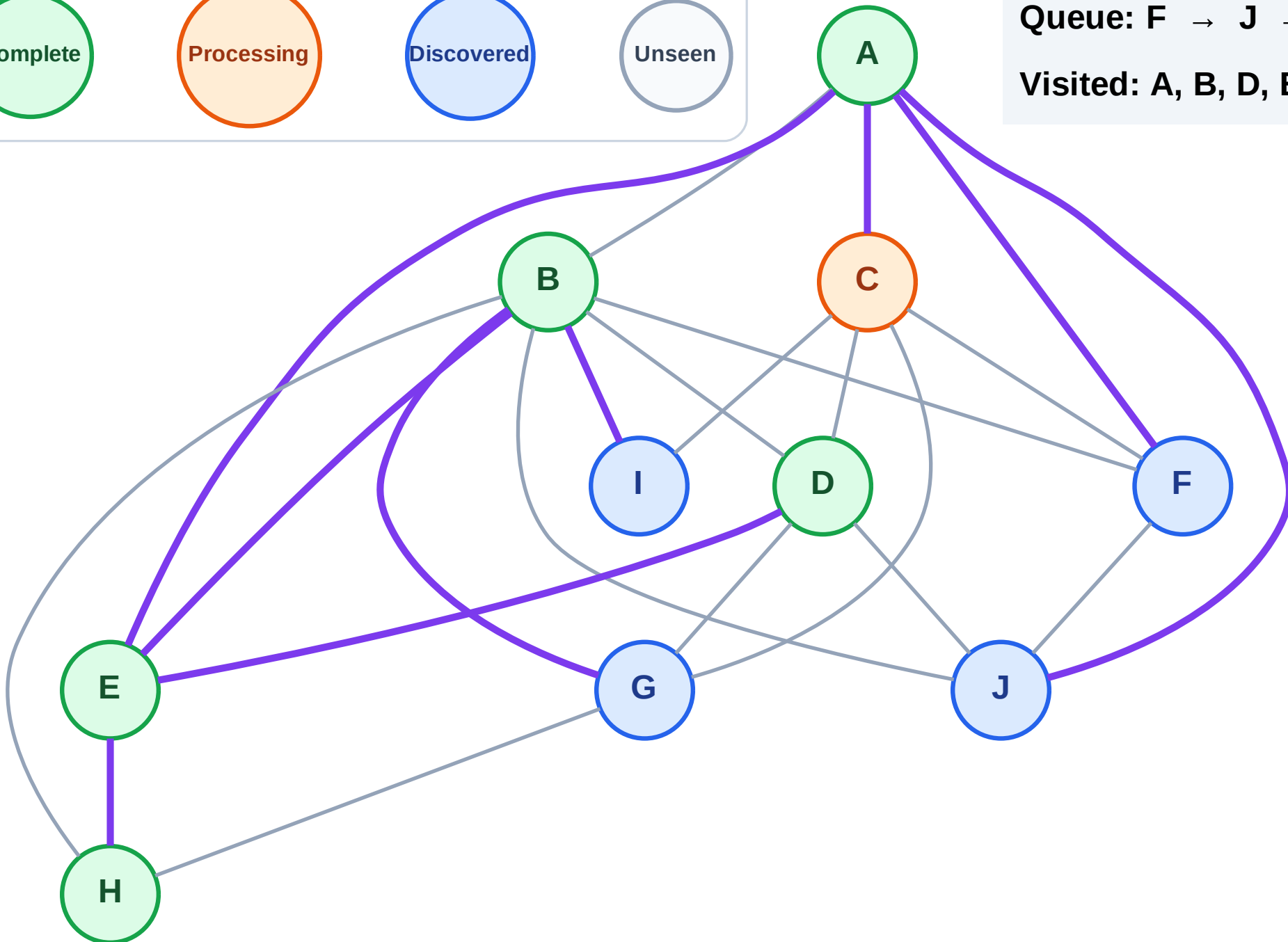
Discovered



Unseen

Queue: F → J → G → I

Visited: A, B, D, E, H



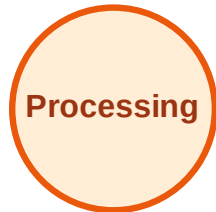
BFS Traversal

Step 13: No new neighbors from C

Legend



Complete



Processing



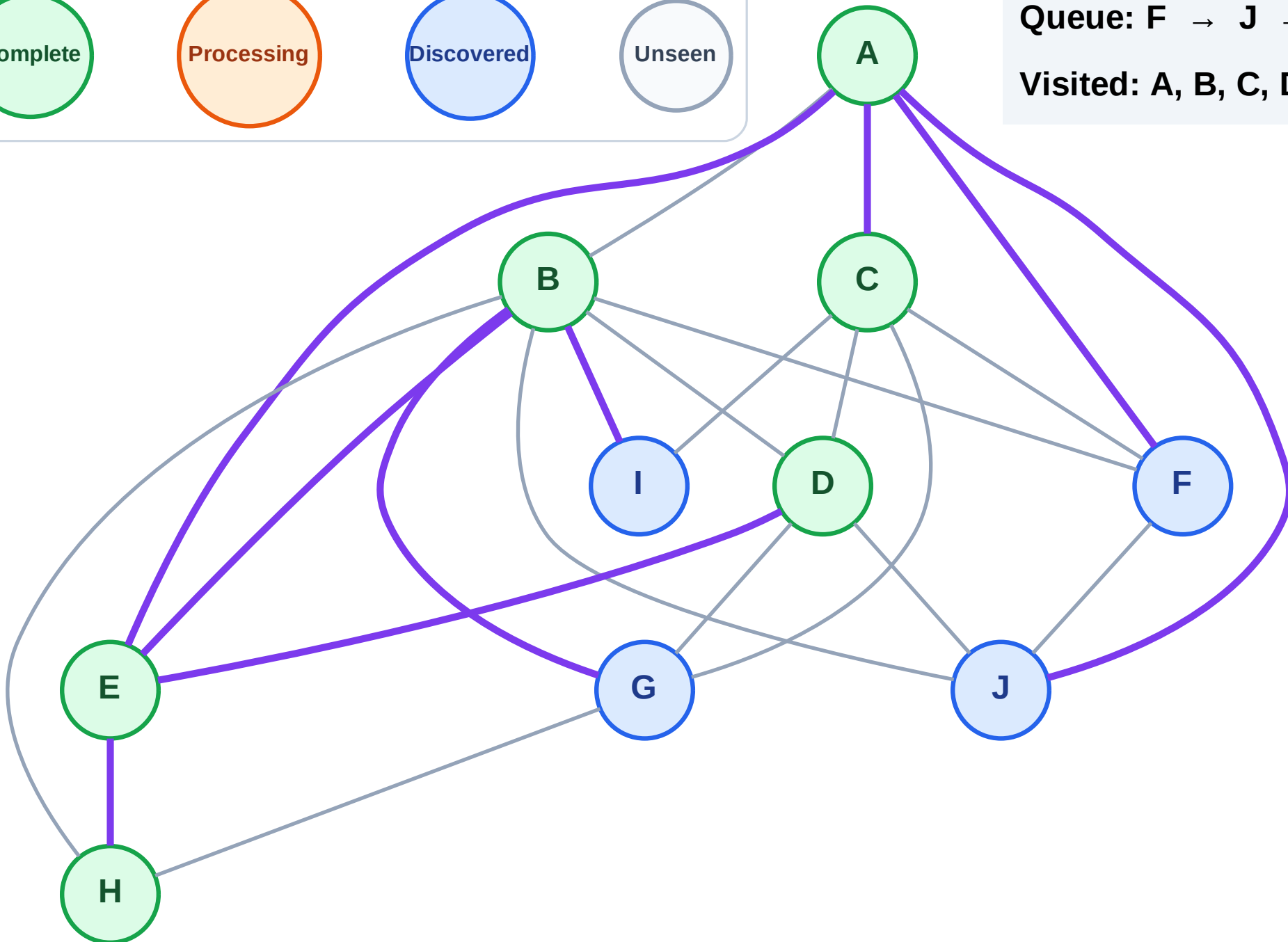
Discovered



Unseen

Queue: F → J → G → I

Visited: A, B, C, D, E, H



BFS Traversal

Step 14: Process node F

Legend

Complete

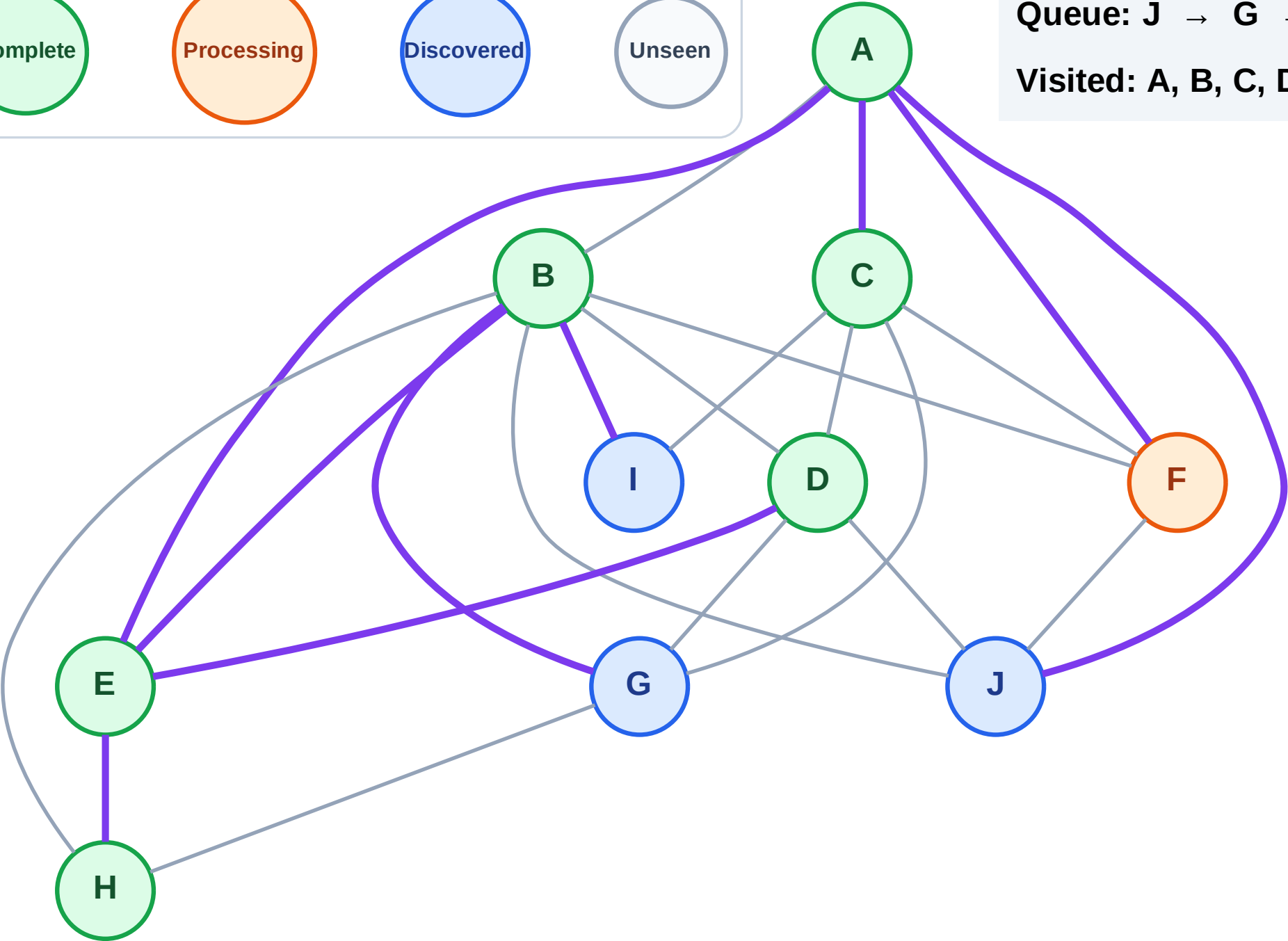
Processing

Discovered

Unseen

Queue: J → G → I

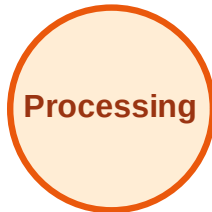
Visited: A, B, C, D, E, H



BFS Traversal

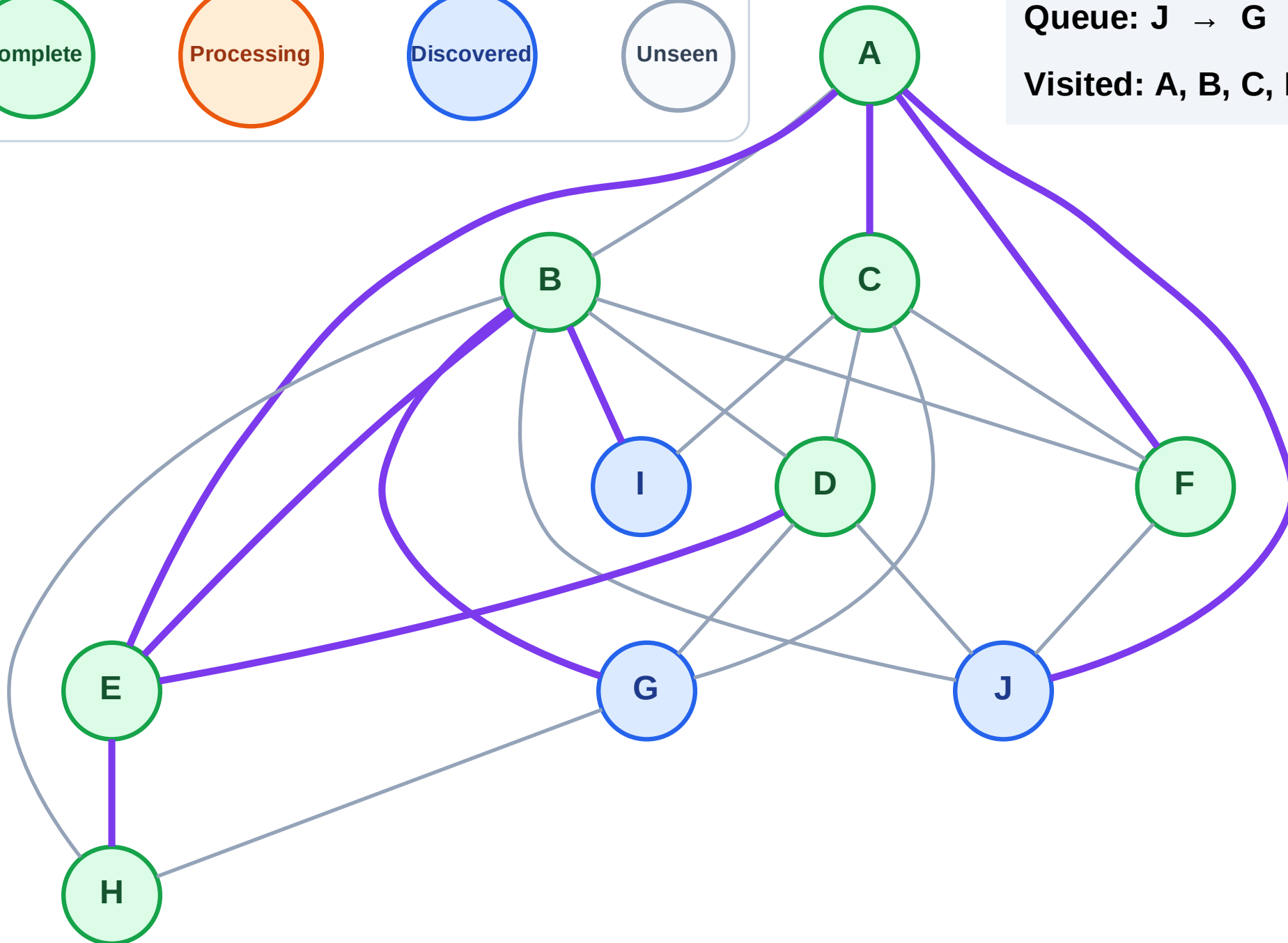
Step 15: No new neighbors from F

Legend



Queue: J → G → I

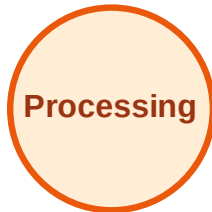
Visited: A, B, C, D, E, F, H



BFS Traversal

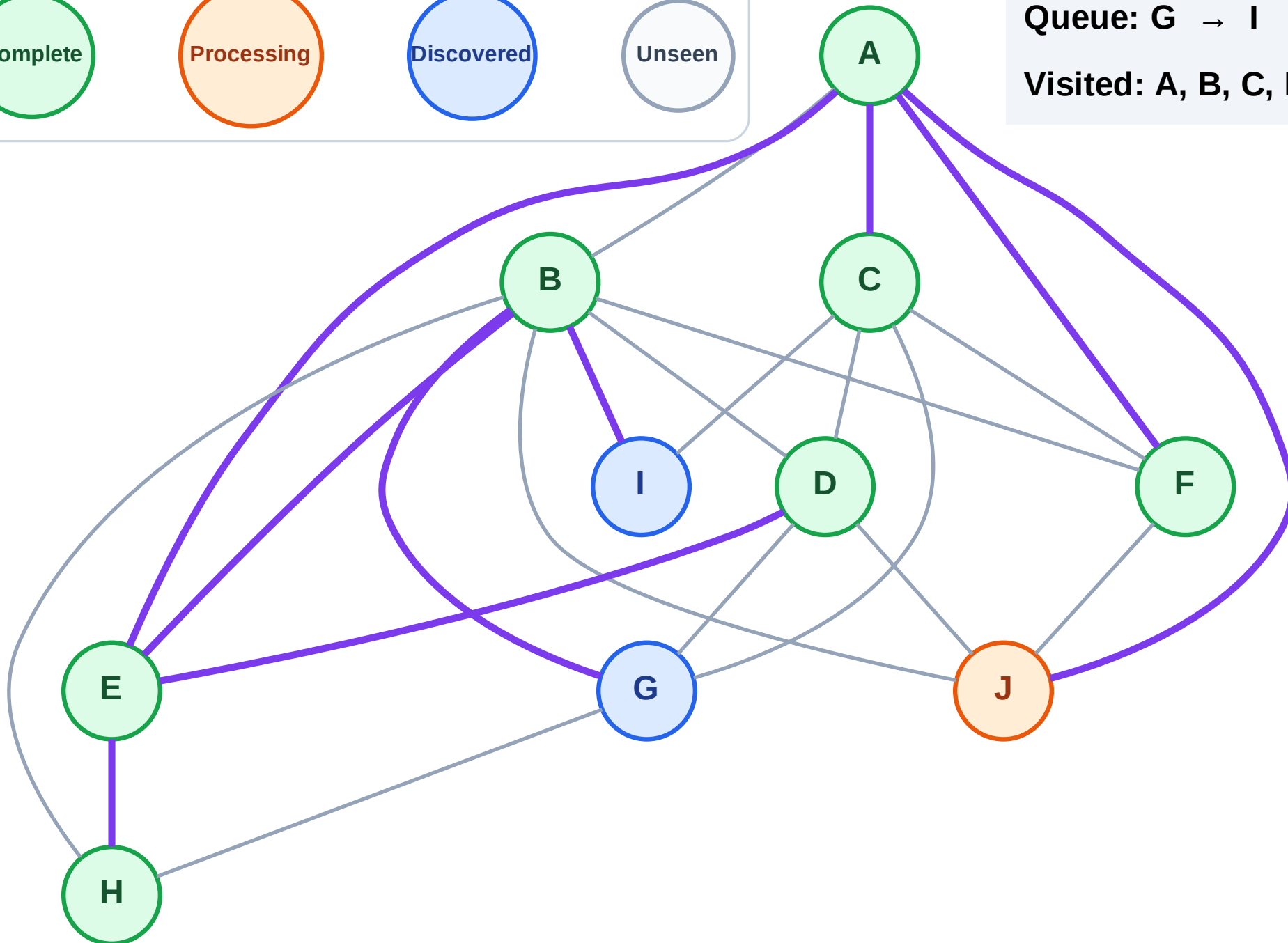
Step 16: Process node J

Legend



Queue: G → I

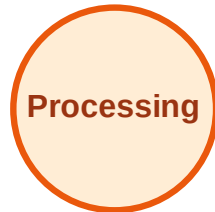
Visited: A, B, C, D, E, F, H



BFS Traversal

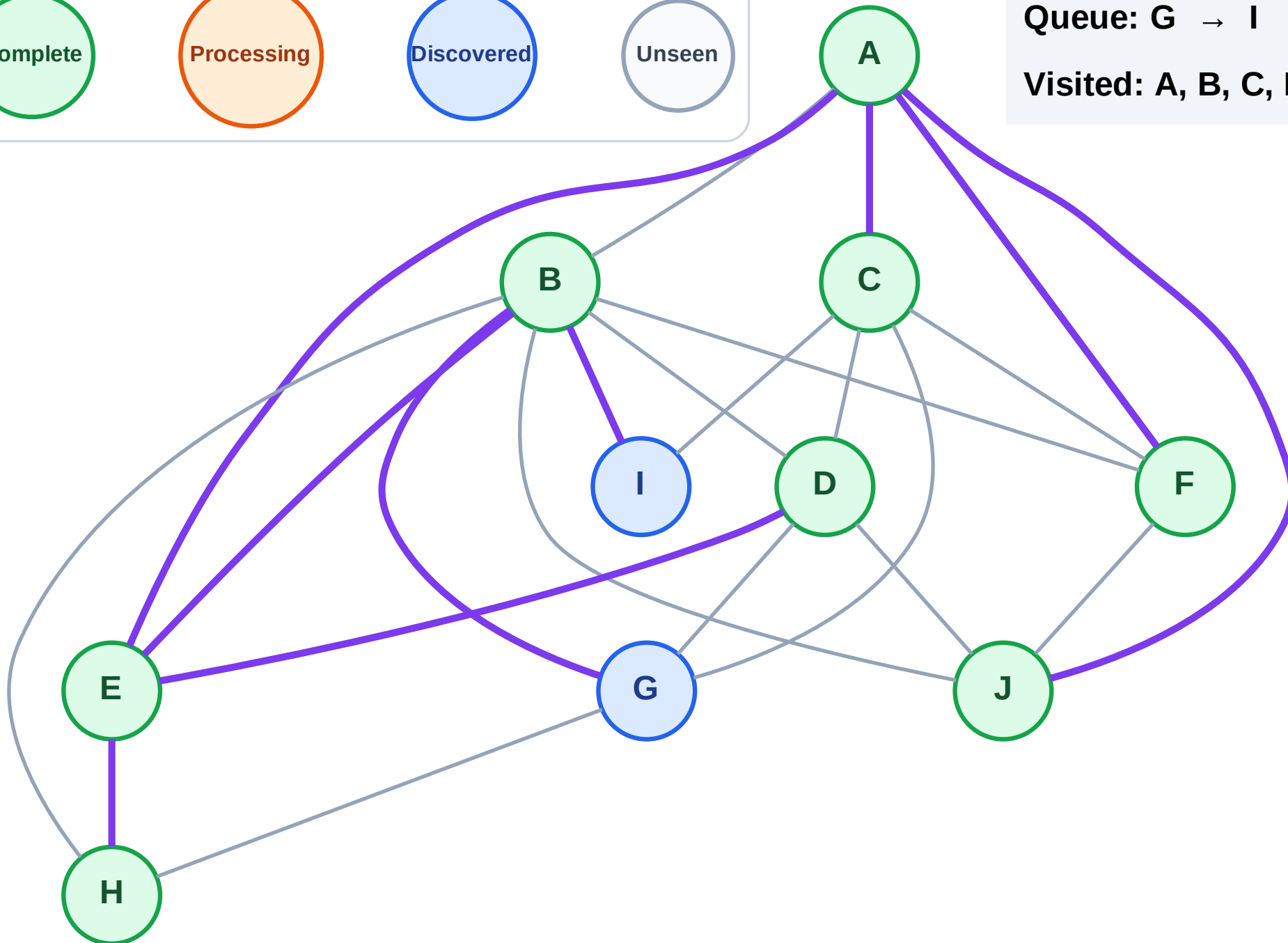
Step 17: No new neighbors from J

Legend



Queue: G → I

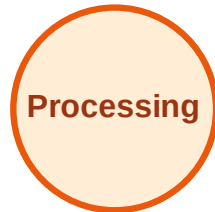
Visited: A, B, C, D, E, F, H, J



BFS Traversal

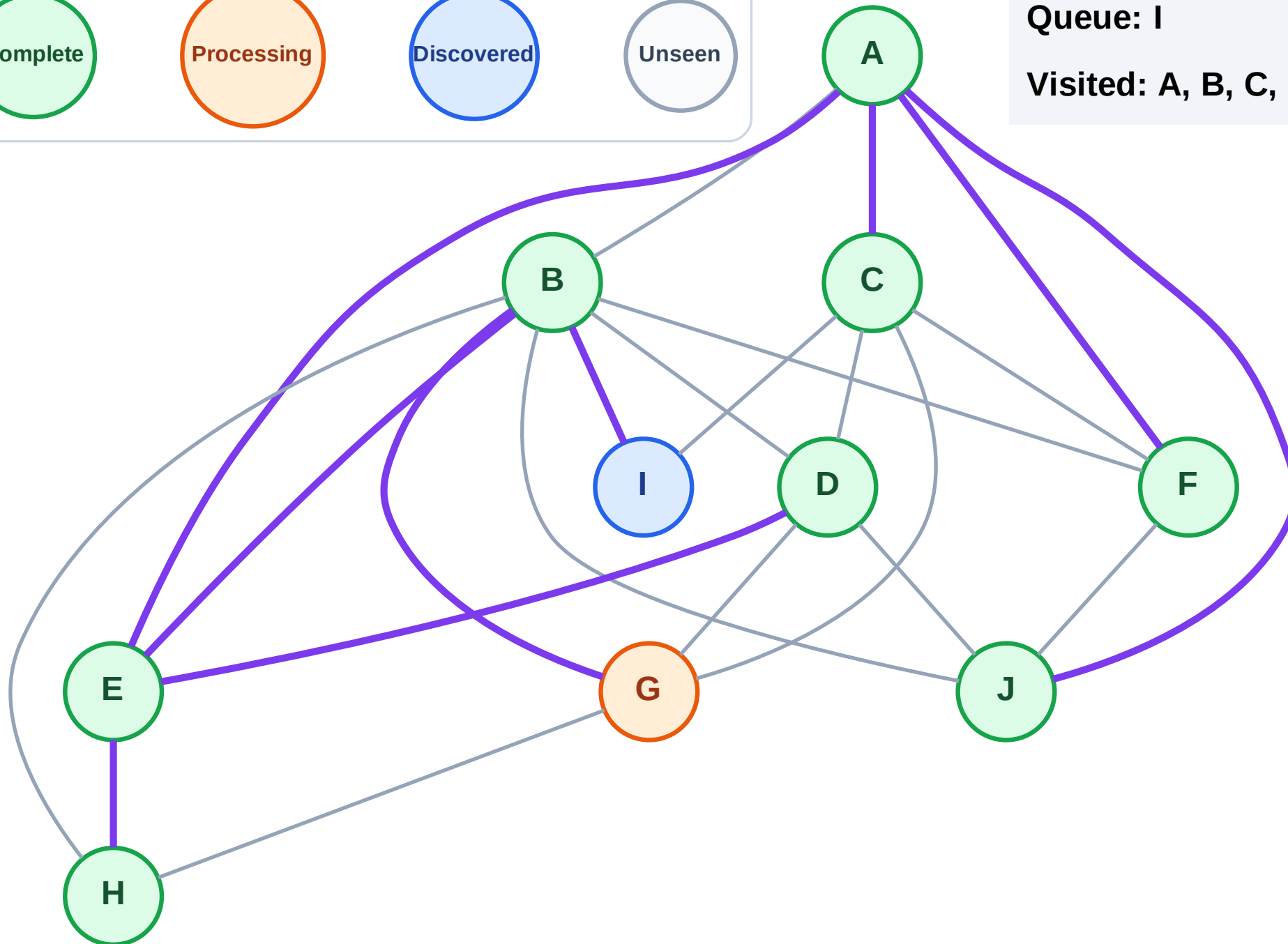
Step 18: Process node G

Legend



Queue: I

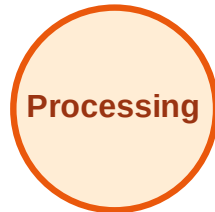
Visited: A, B, C, D, E, F, H, J



BFS Traversal

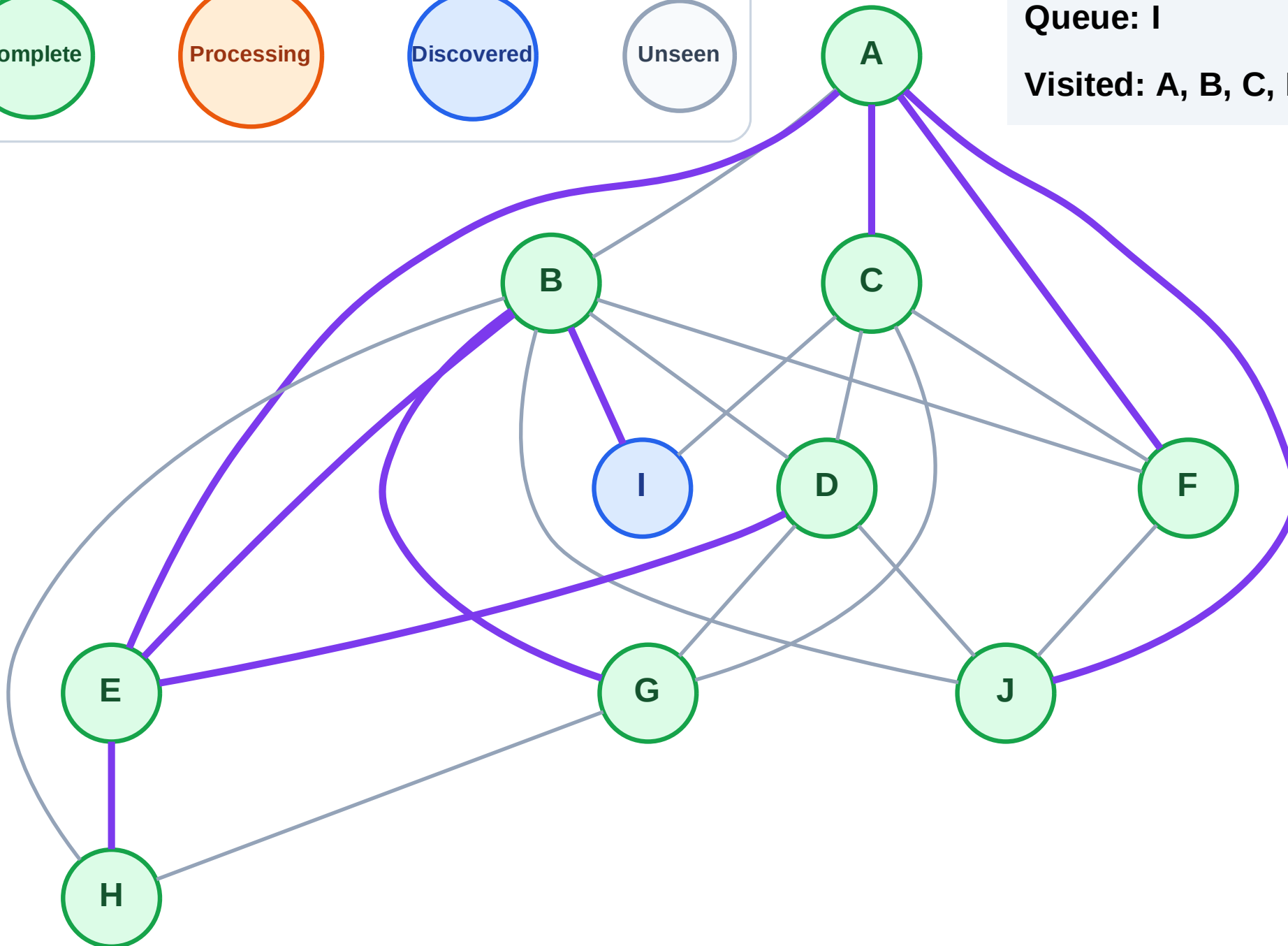
Step 19: No new neighbors from G

Legend



Queue: I

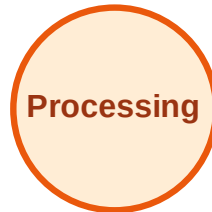
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

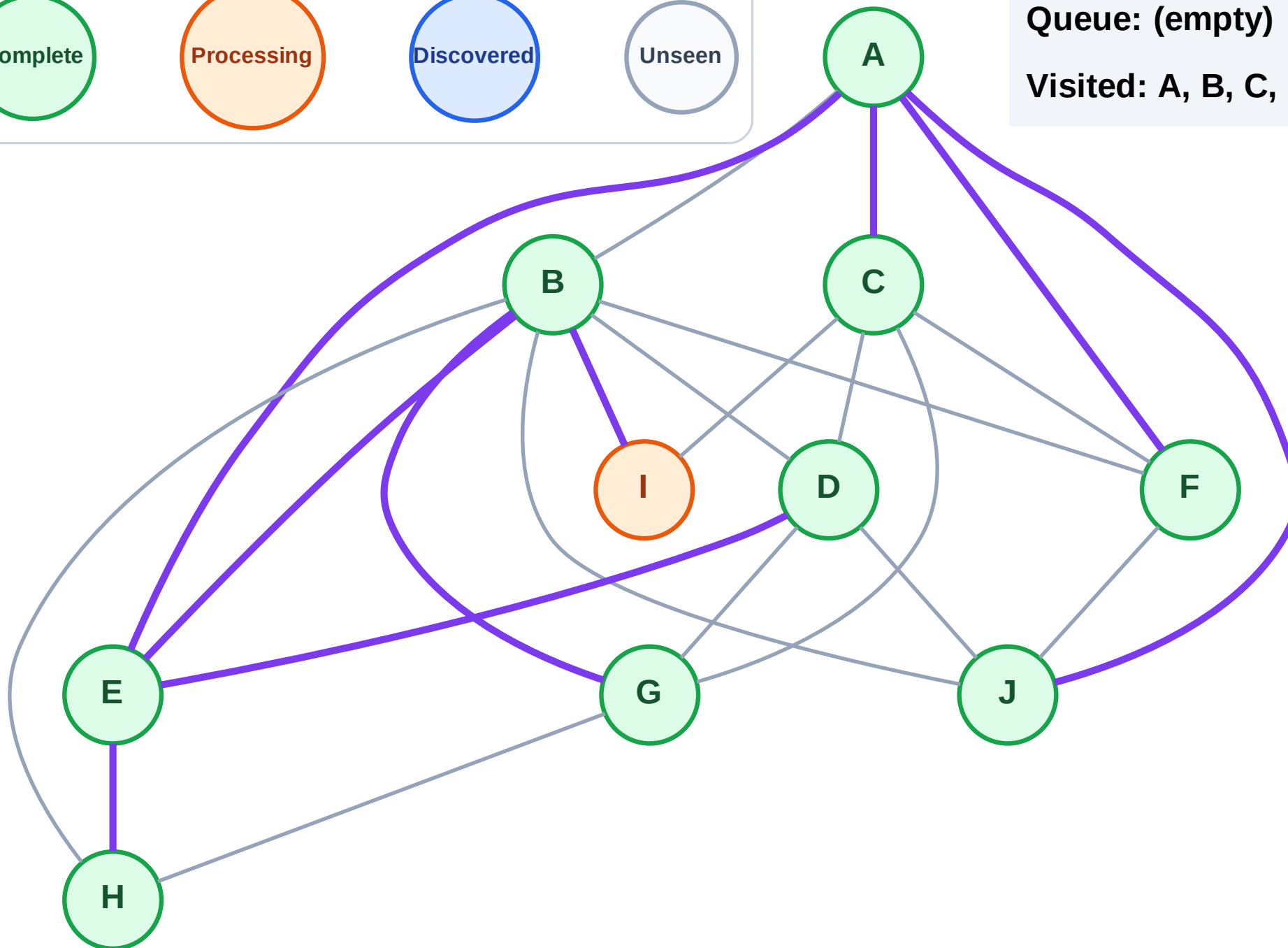
Step 20: Process node I

Legend



Queue: (empty)

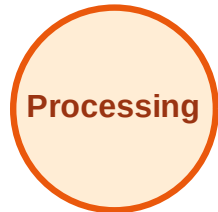
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

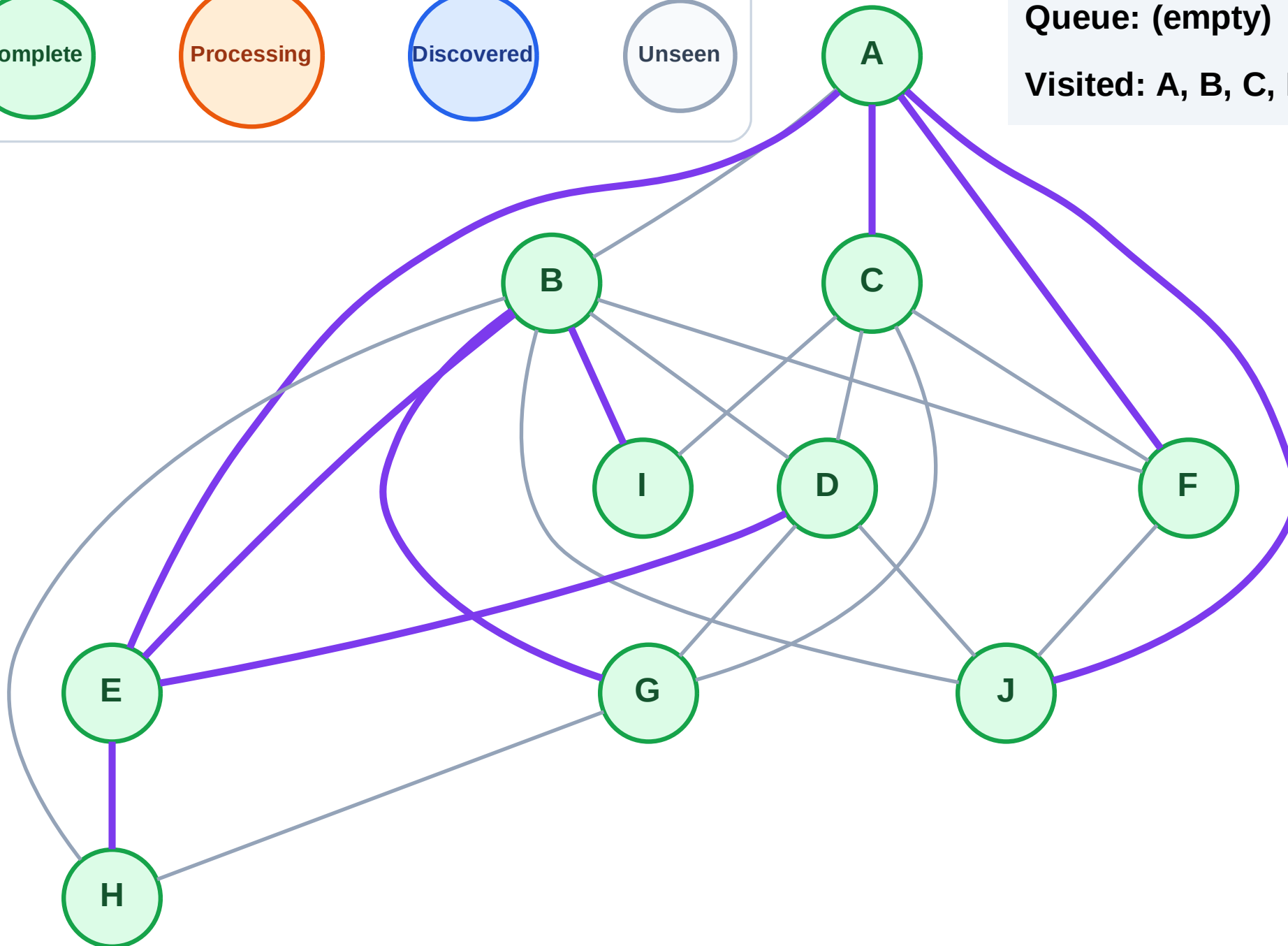
Step 21: No new neighbors from I

Legend



Queue: (empty)

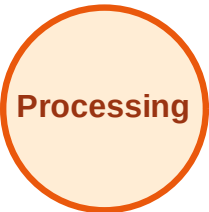
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

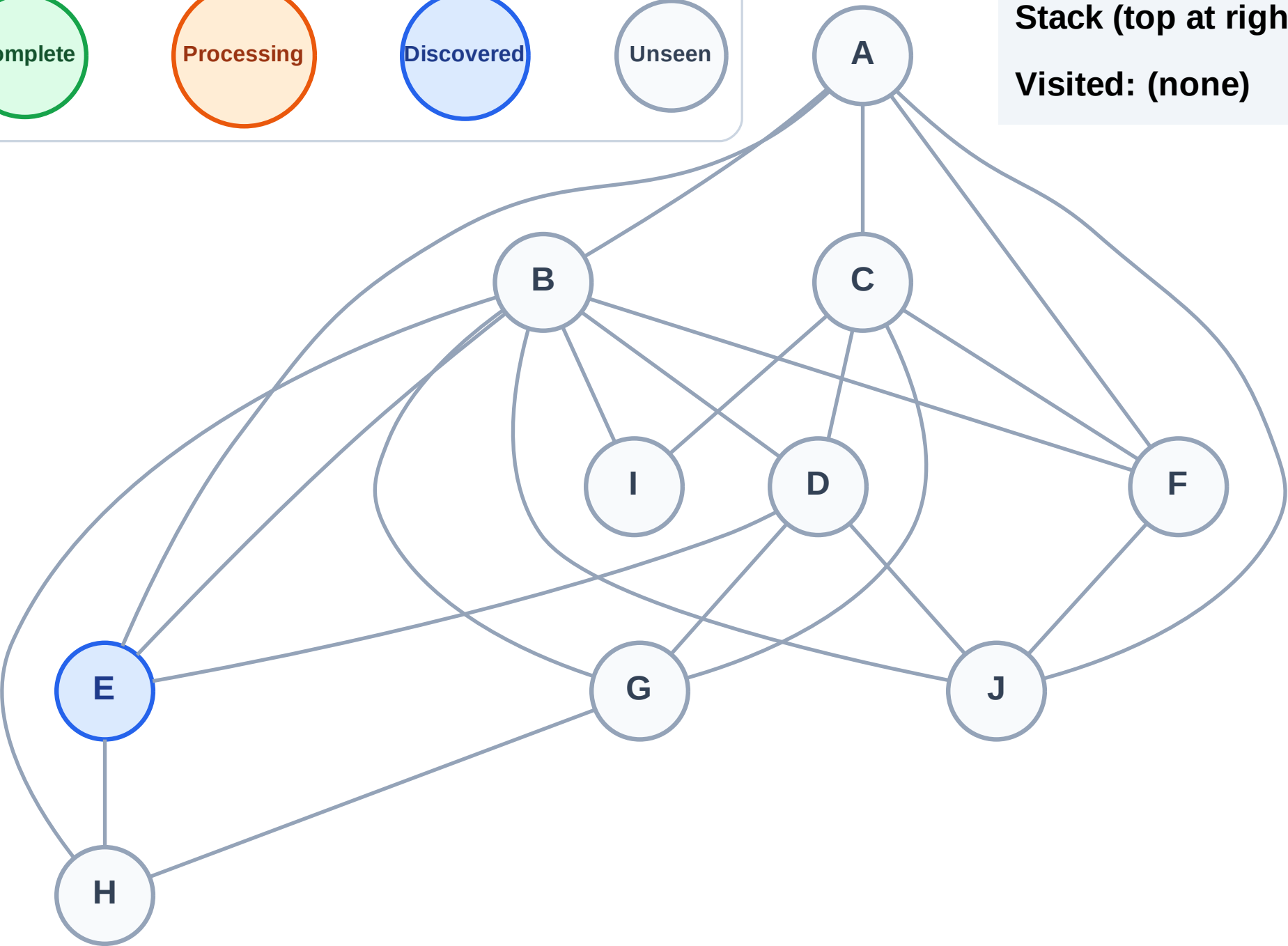
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

Complete

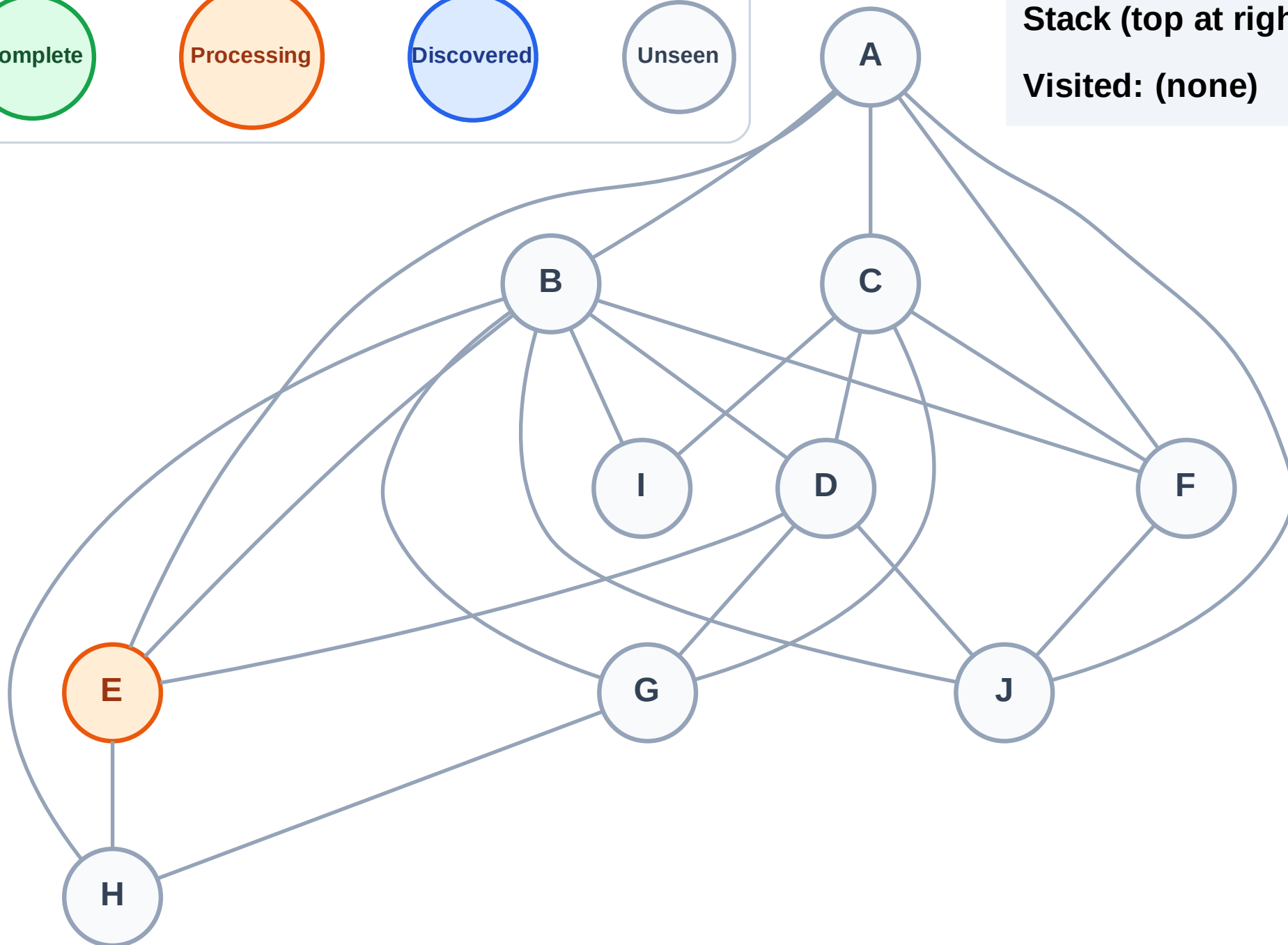
Processing

Discovered

Unseen

Stack (top at right): (empty)

Visited: (none)



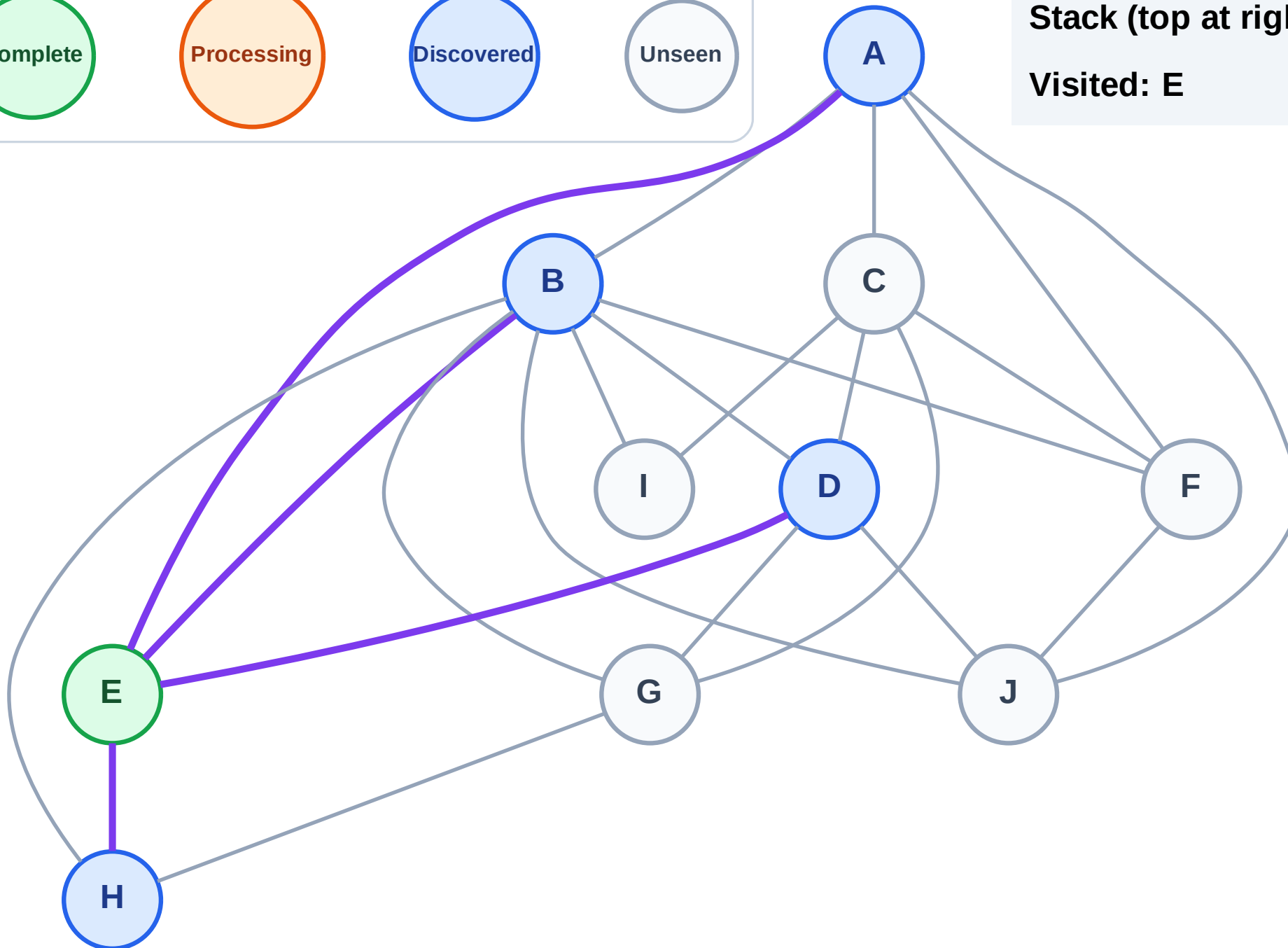
DFS Traversal

Step 3: Discover H, D, B, A from E

Legend



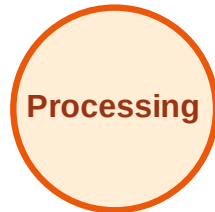
Stack (top at right): H | D | B | A
Visited: E



DFS Traversal

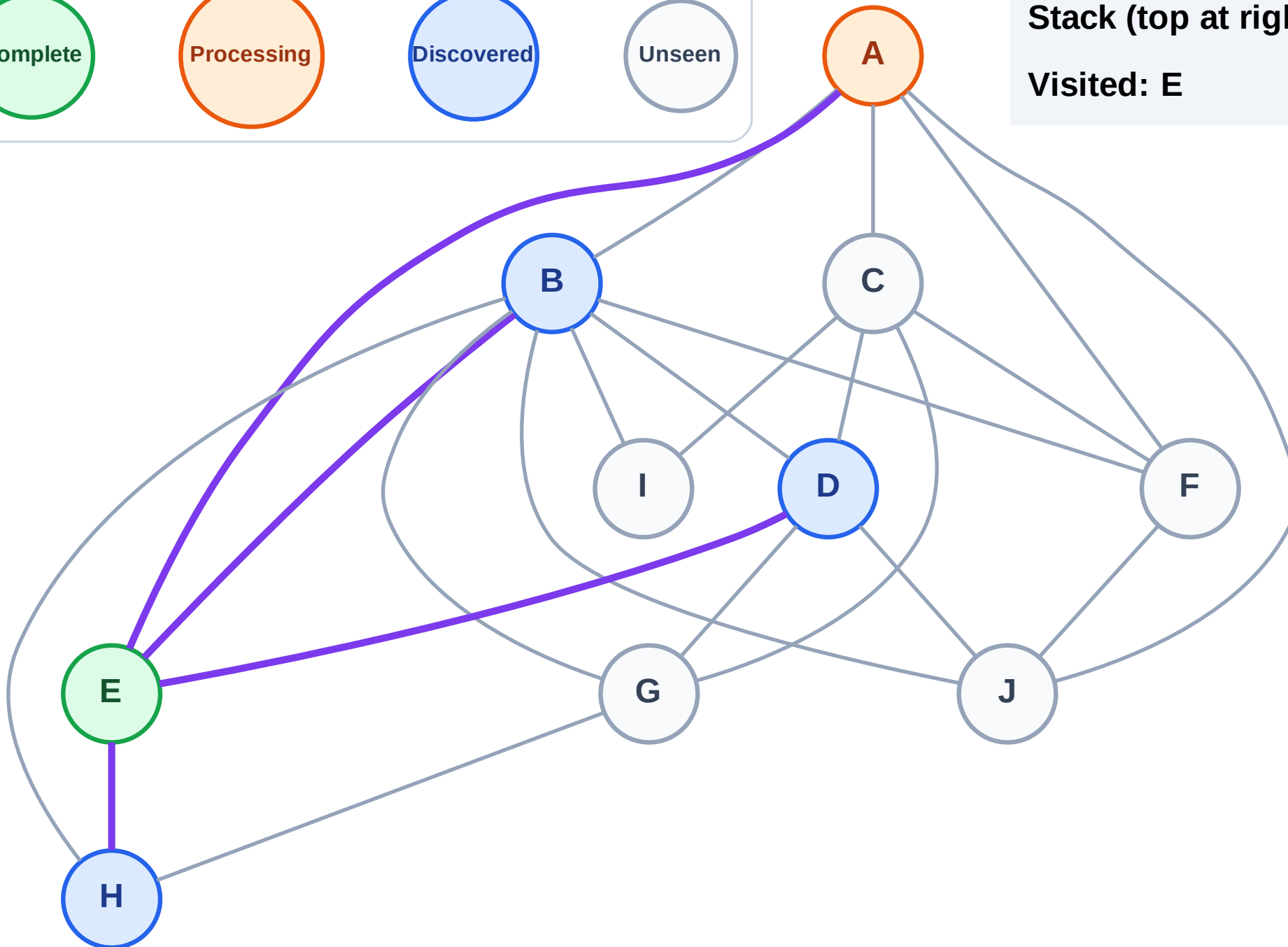
Step 4: Process node A

Legend



Stack (top at right): H | D | B

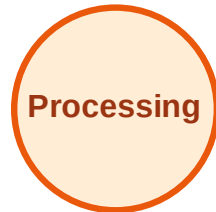
Visited: E



DFS Traversal

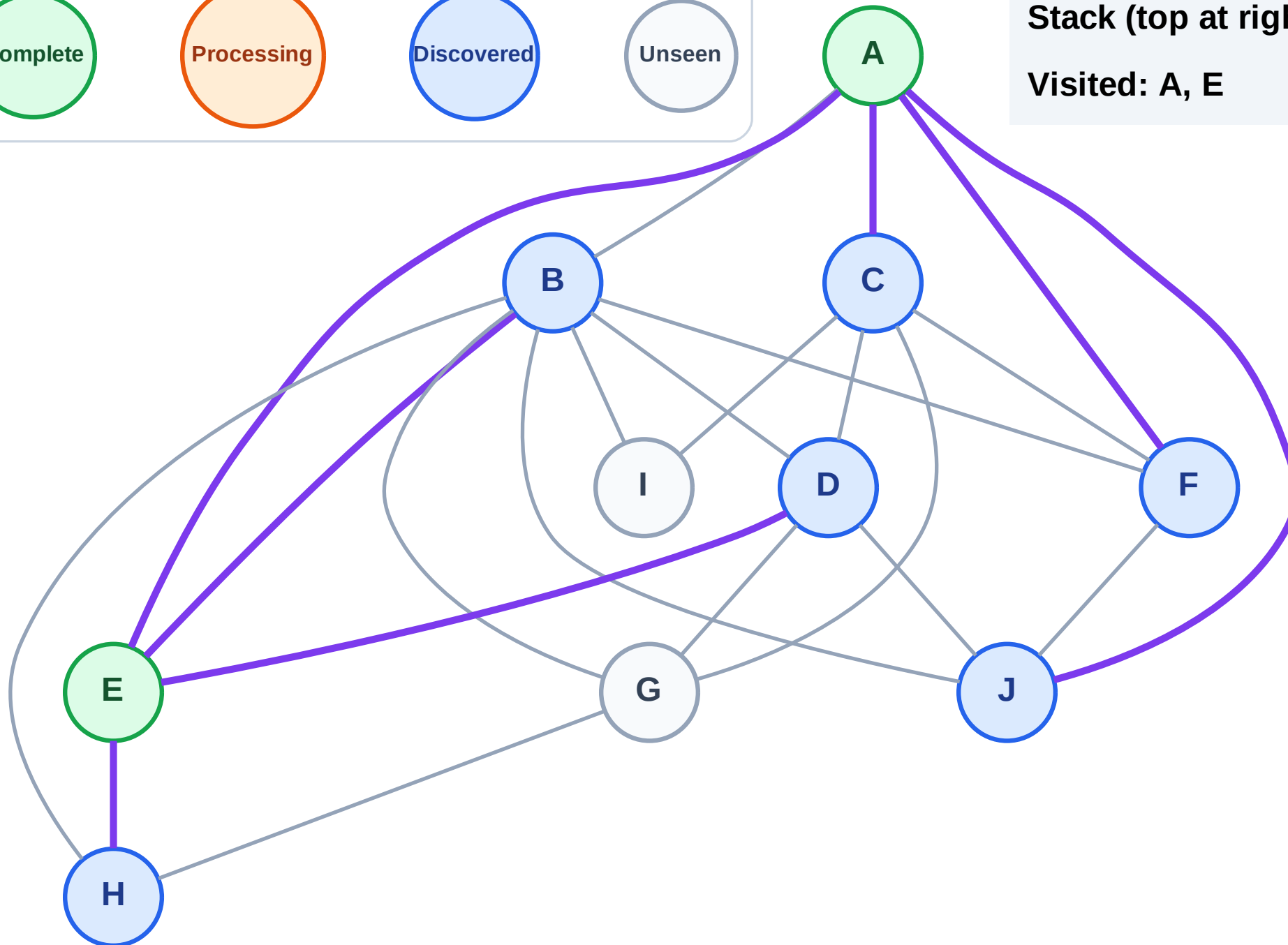
Step 5: Discover J, F, C from A

Legend



Stack (top at right): H | D | B | J | F | C

Visited: A, E



DFS Traversal

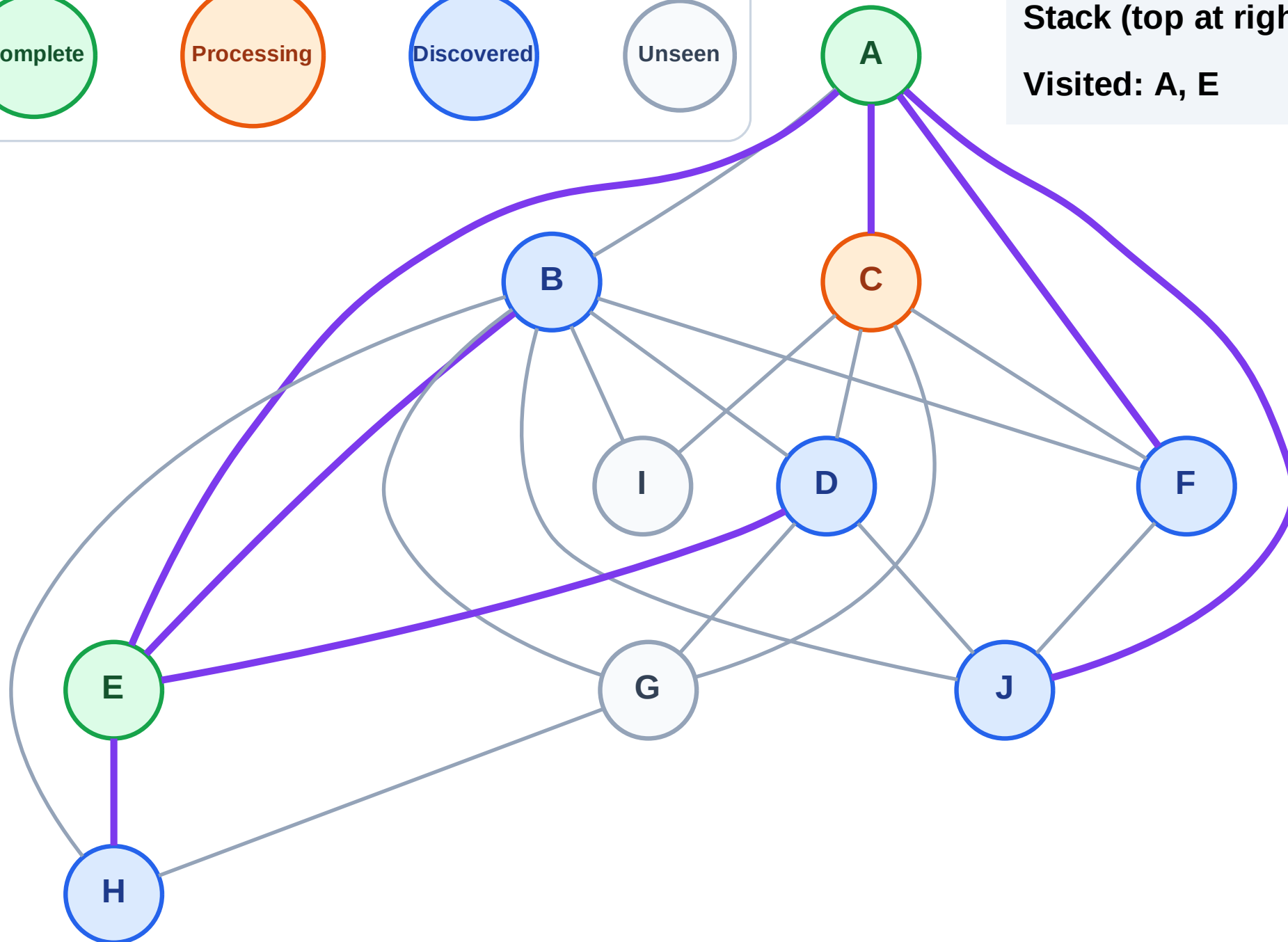
Step 6: Process node C

Legend



Stack (top at right): H | D | B | J | F

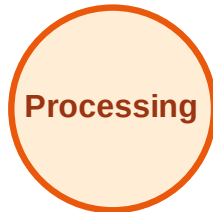
Visited: A, E



DFS Traversal

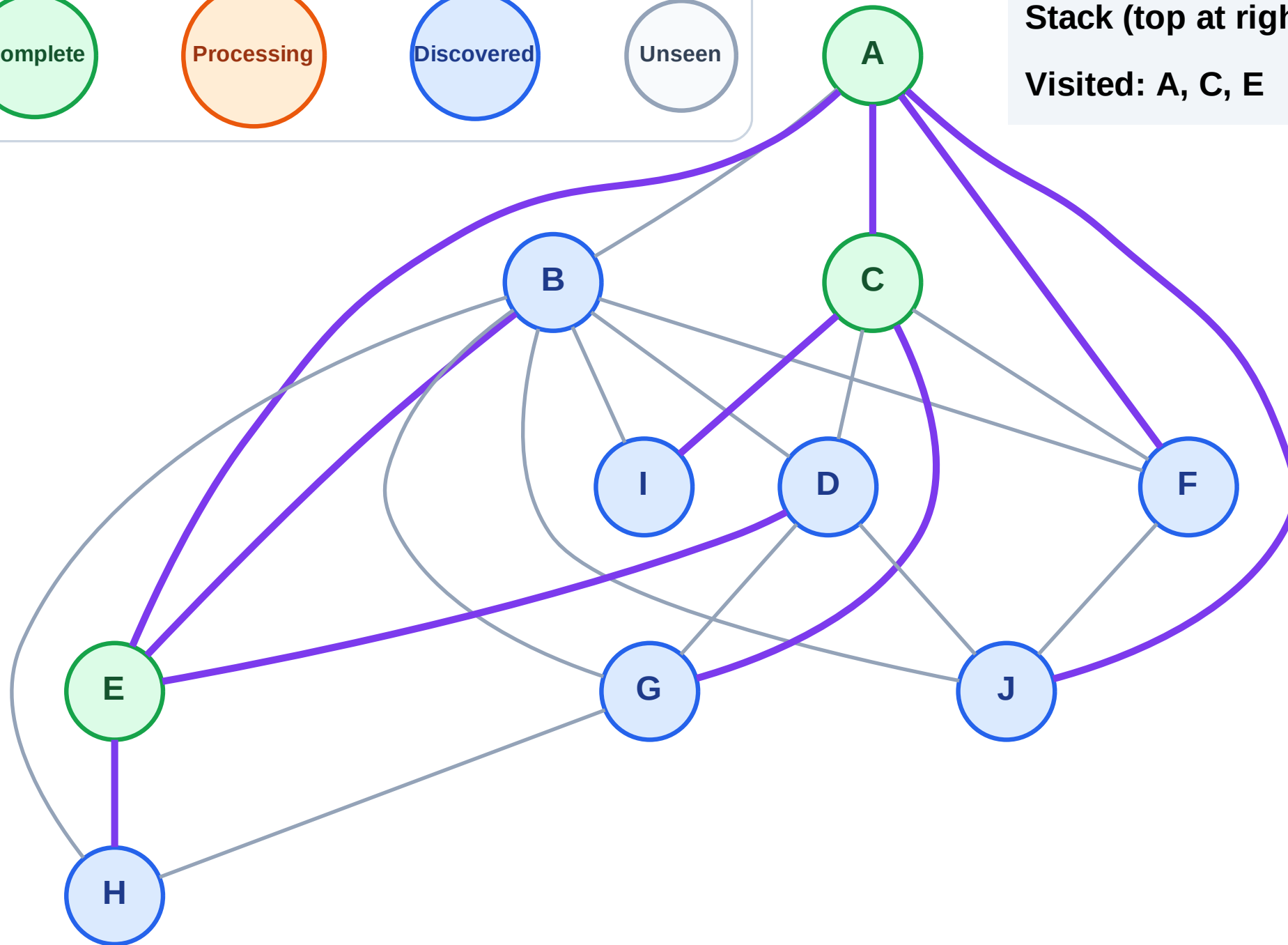
Step 7: Discover I, G from C

Legend



Stack (top at right): H | D | B | J | F | I | G

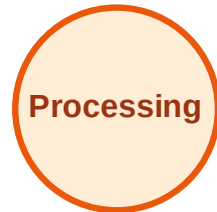
Visited: A, C, E



DFS Traversal

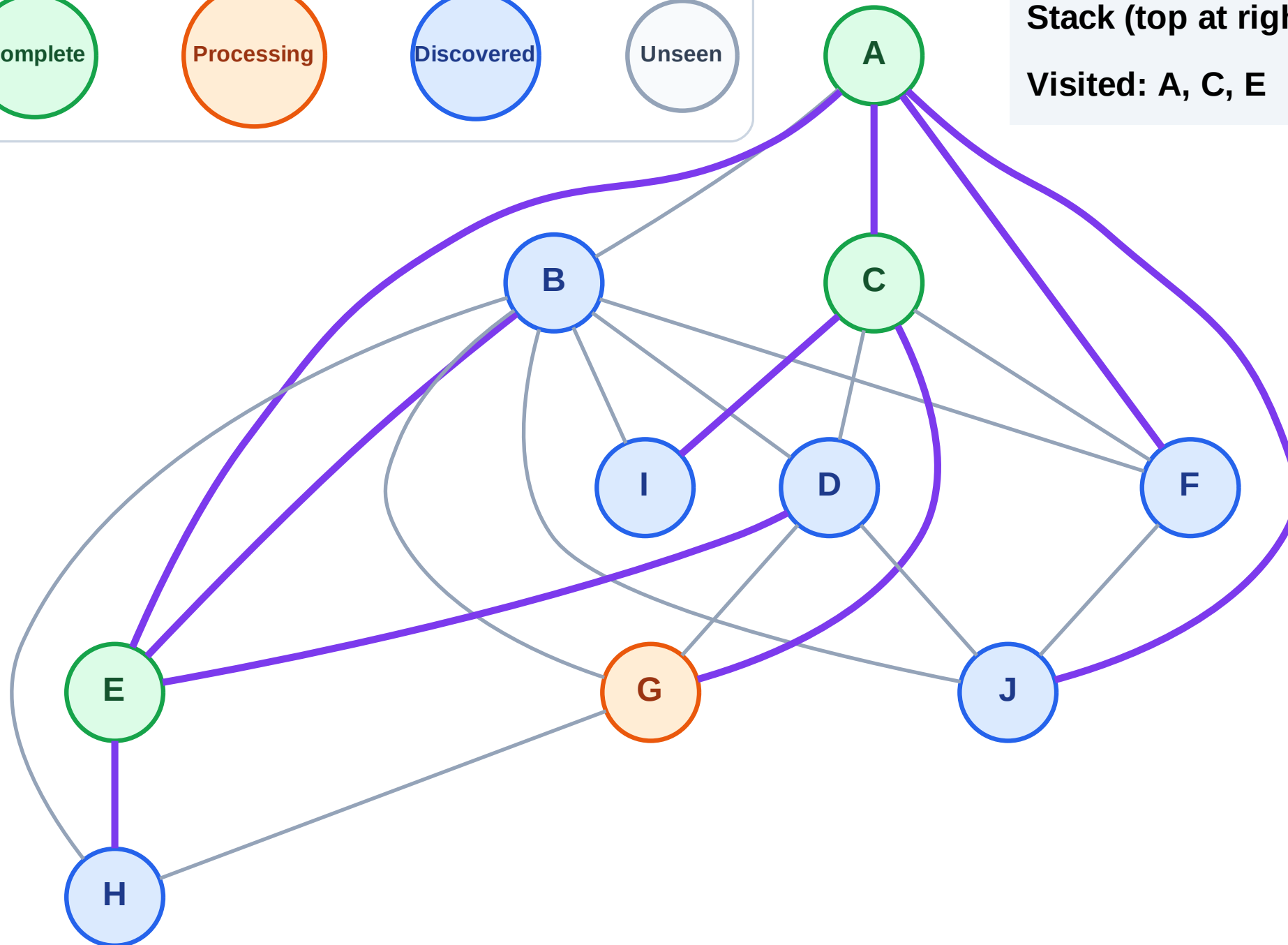
Step 8: Process node G

Legend



Stack (top at right): H | D | B | J | F | I

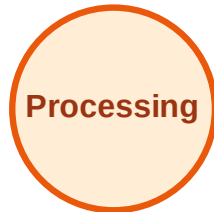
Visited: A, C, E



DFS Traversal

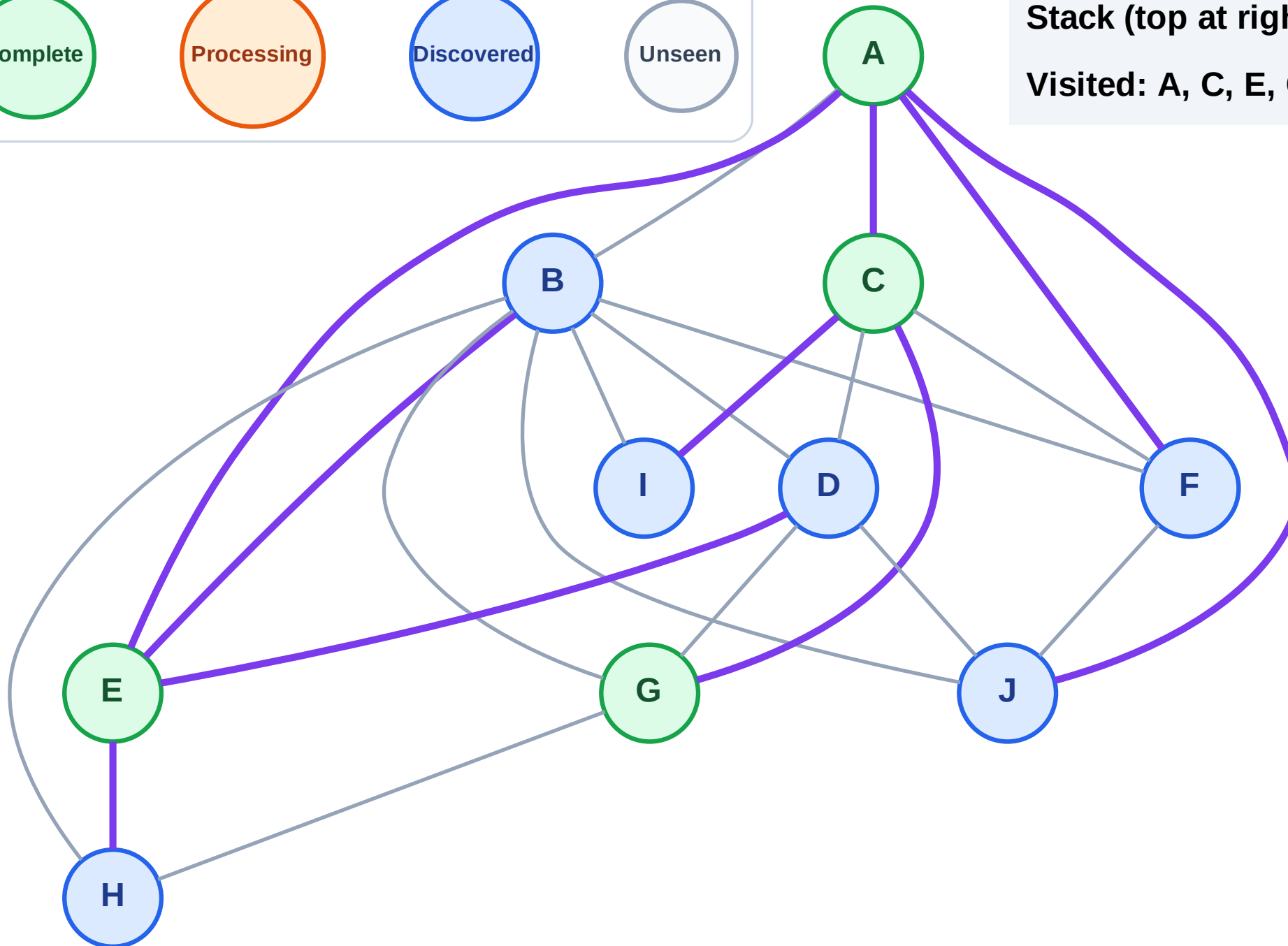
Step 9: No new neighbors from G

Legend



Stack (top at right): H | D | B | J | F | I

Visited: A, C, E, G



DFS Traversal

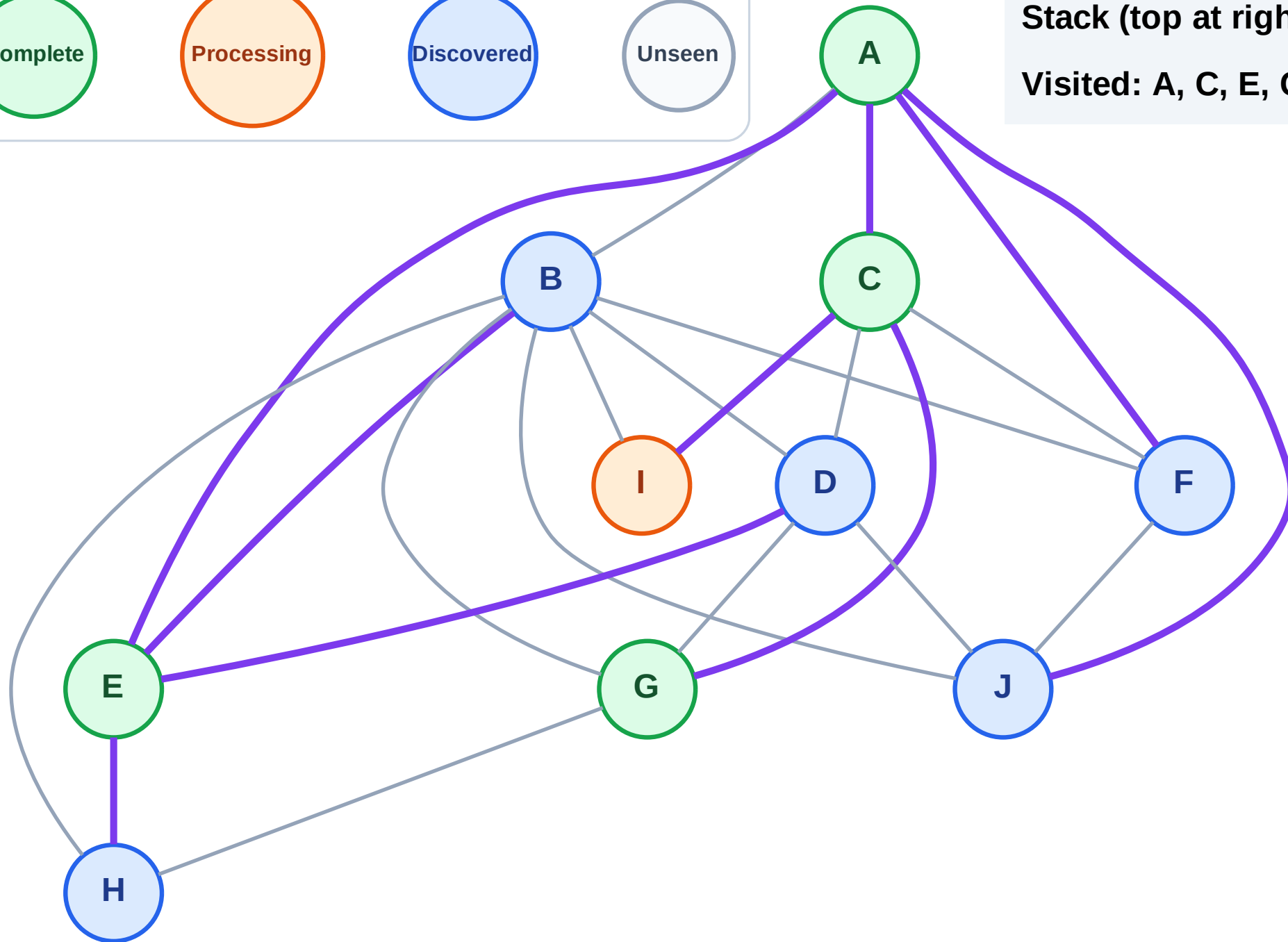
Step 10: Process node I

Legend



Stack (top at right): H | D | B | J | F

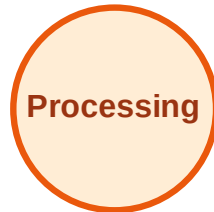
Visited: A, C, E, G



DFS Traversal

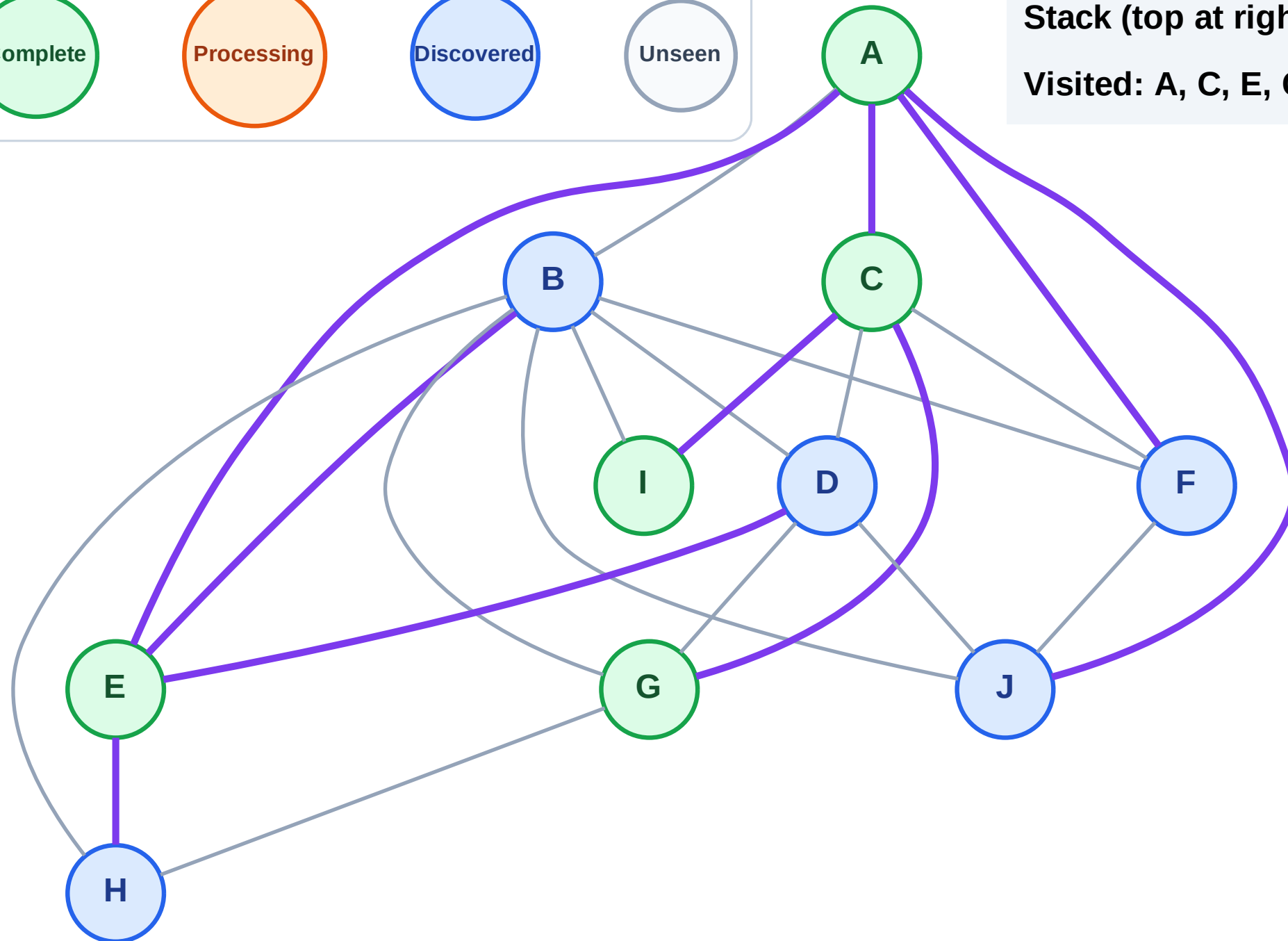
Step 11: No new neighbors from I

Legend



Stack (top at right): H | D | B | J | F

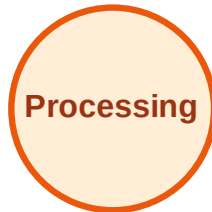
Visited: A, C, E, G, I



DFS Traversal

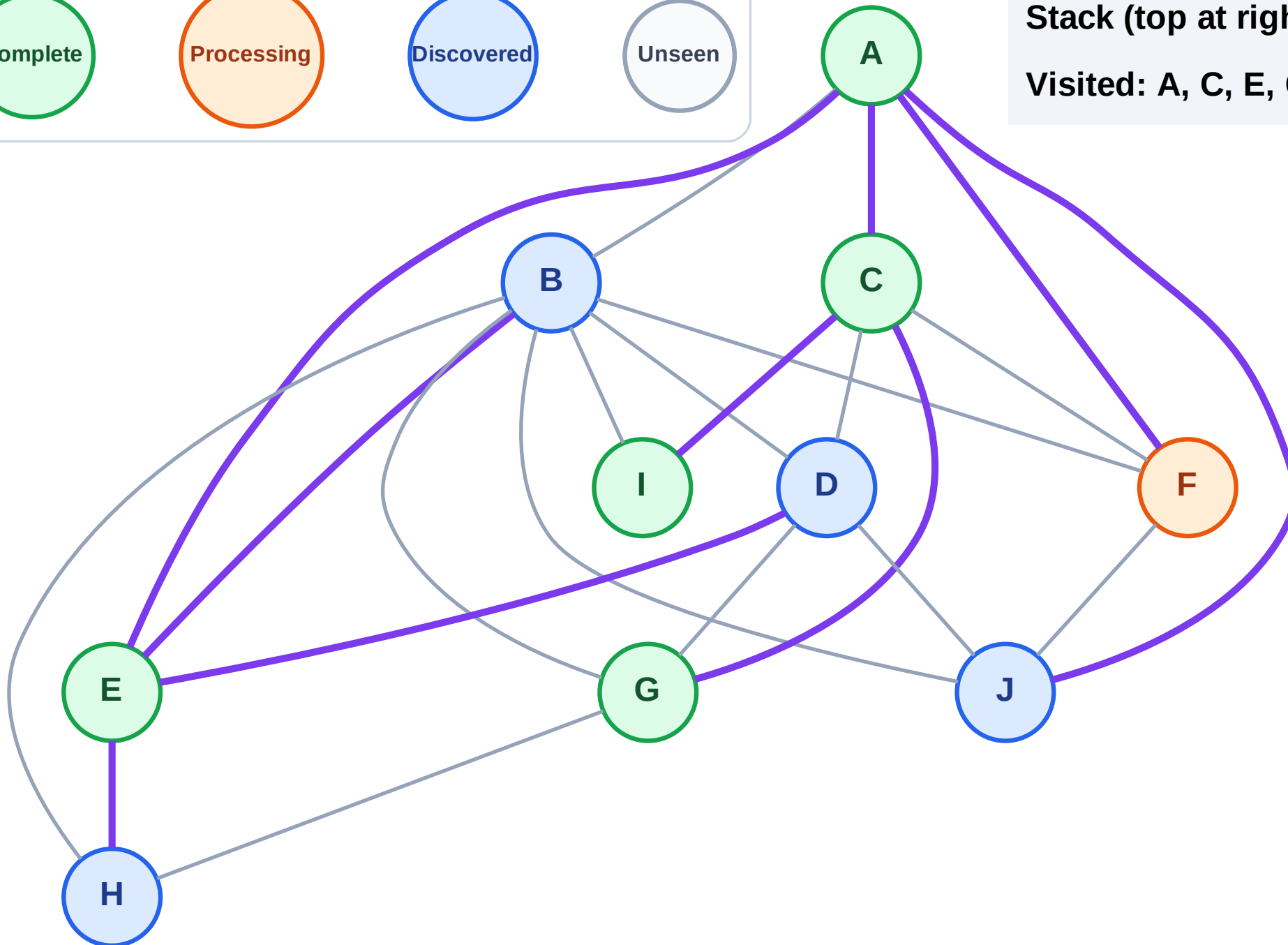
Step 12: Process node F

Legend



Stack (top at right): H | D | B | J

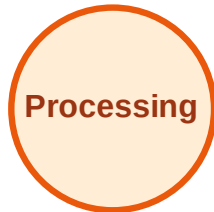
Visited: A, C, E, G, I



DFS Traversal

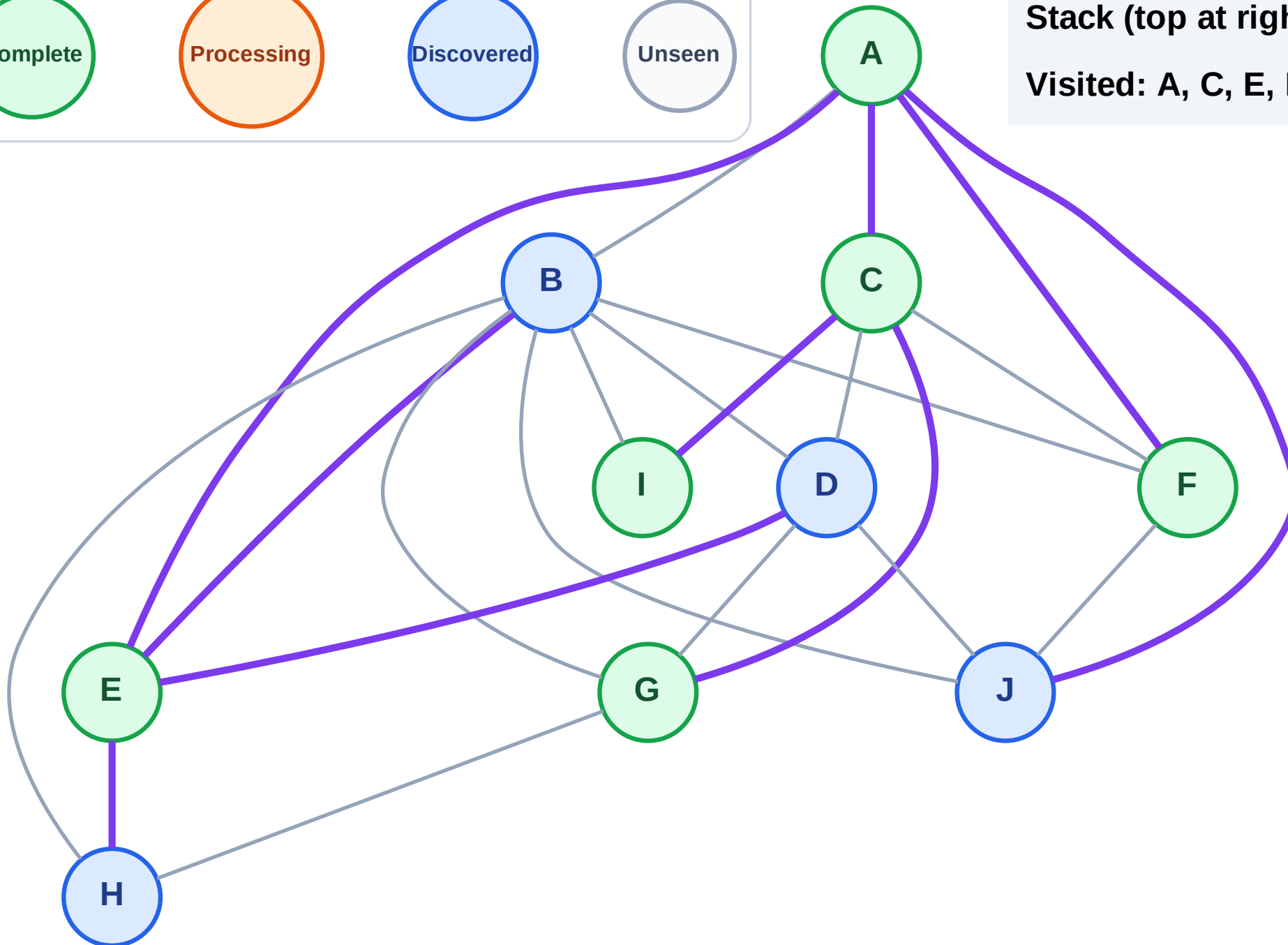
Step 13: No new neighbors from F

Legend



Stack (top at right): H | D | B | J

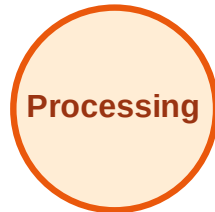
Visited: A, C, E, F, G, I



DFS Traversal

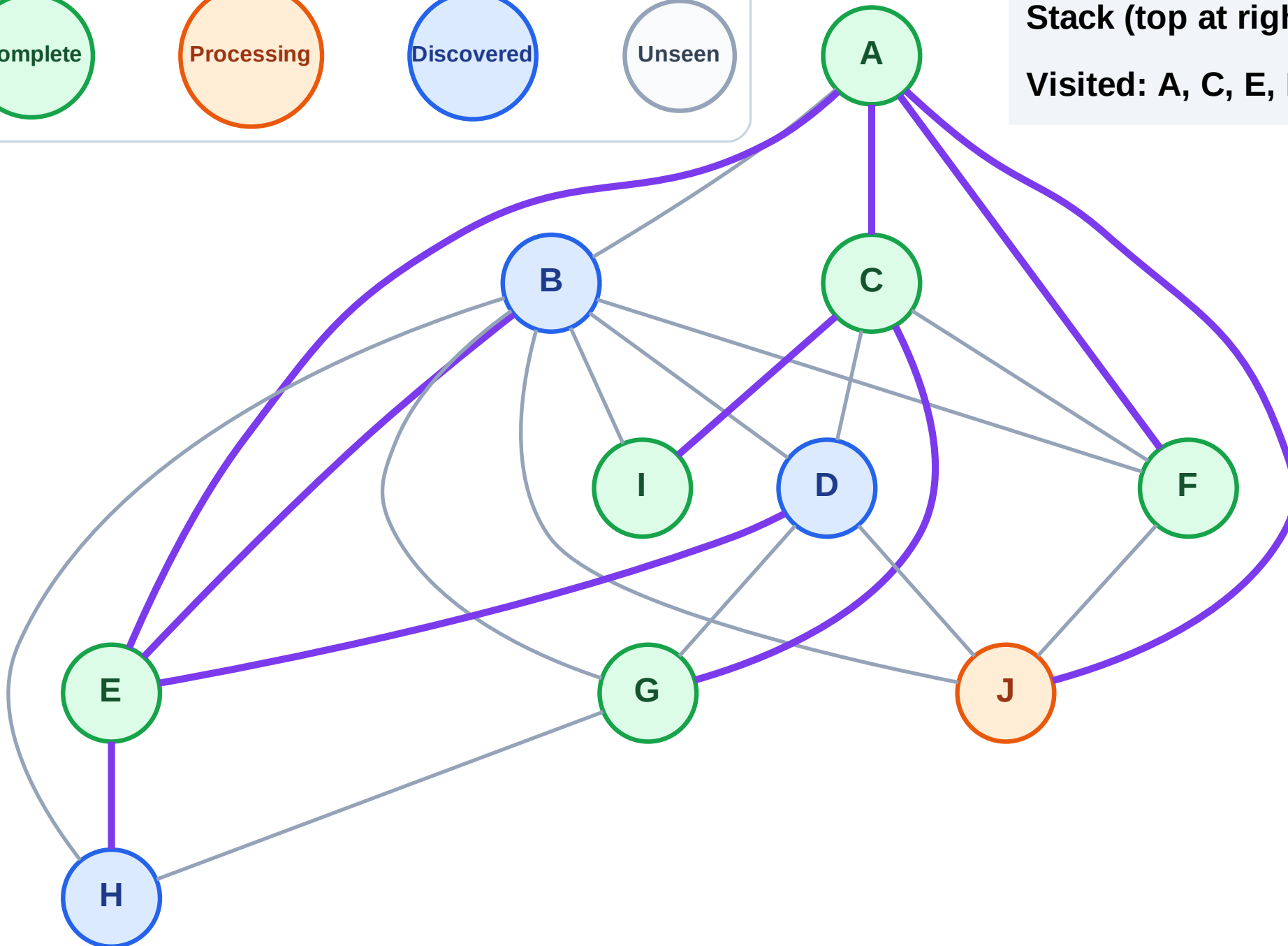
Step 14: Process node J

Legend



Stack (top at right): H | D | B

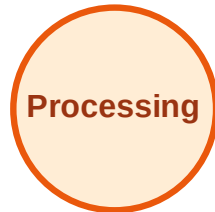
Visited: A, C, E, F, G, I



DFS Traversal

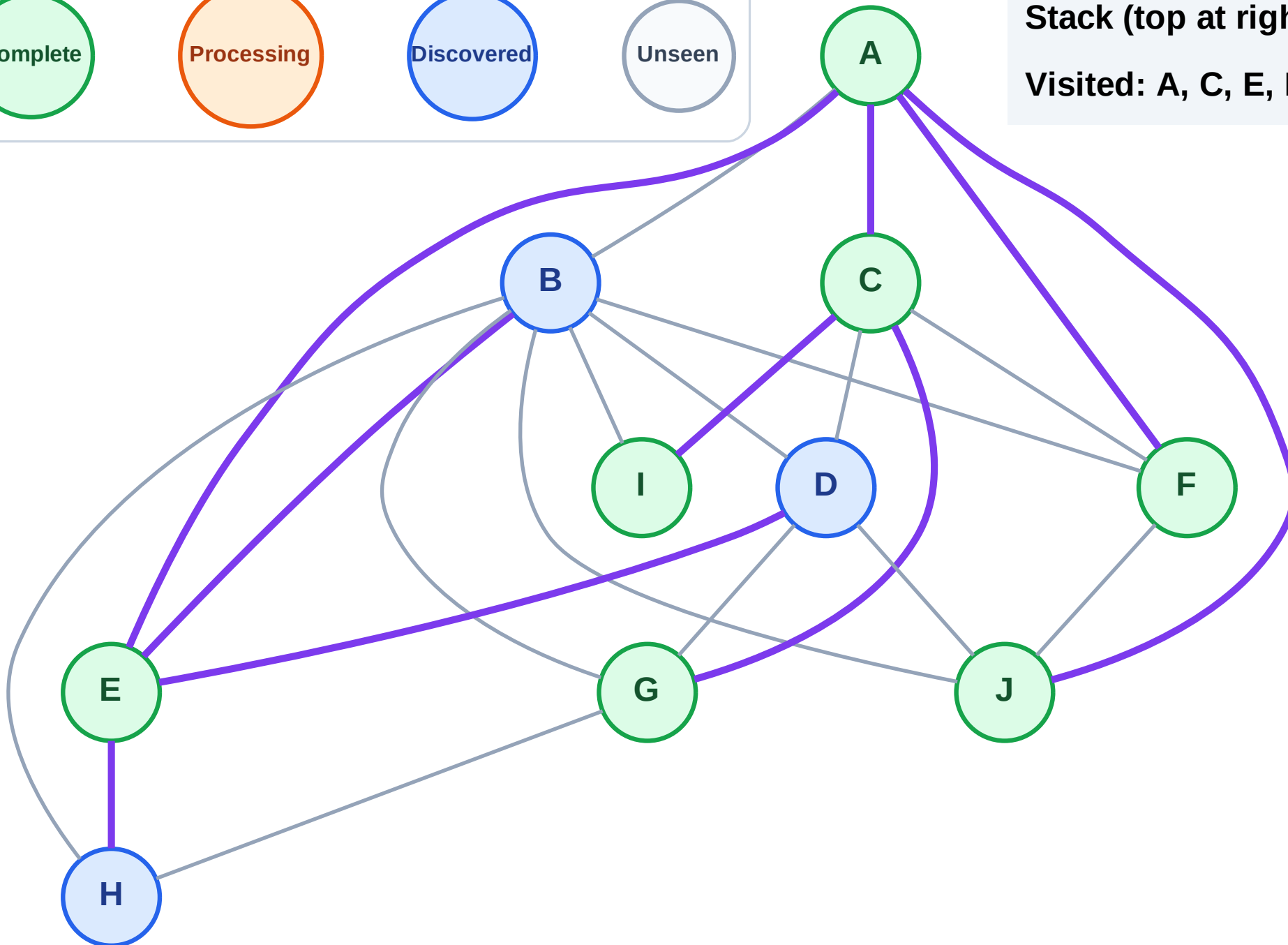
Step 15: No new neighbors from J

Legend



Stack (top at right): H | D | B

Visited: A, C, E, F, G, I, J



DFS Traversal

Step 16: Process node B

Legend

Complete

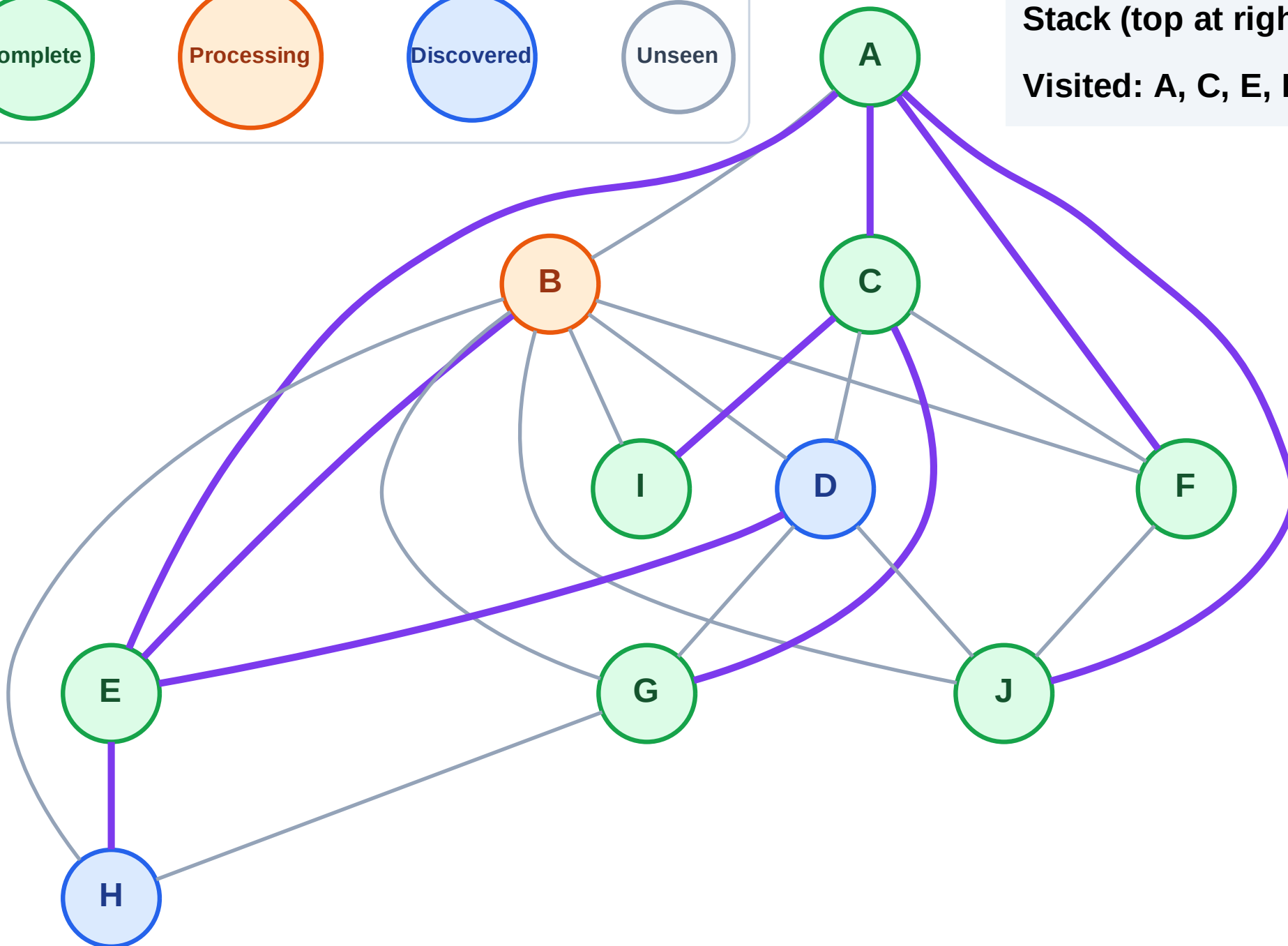
Processing

Discovered

Unseen

Stack (top at right): H | D

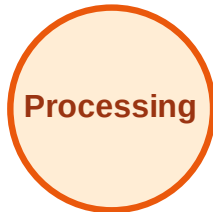
Visited: A, C, E, F, G, I, J



DFS Traversal

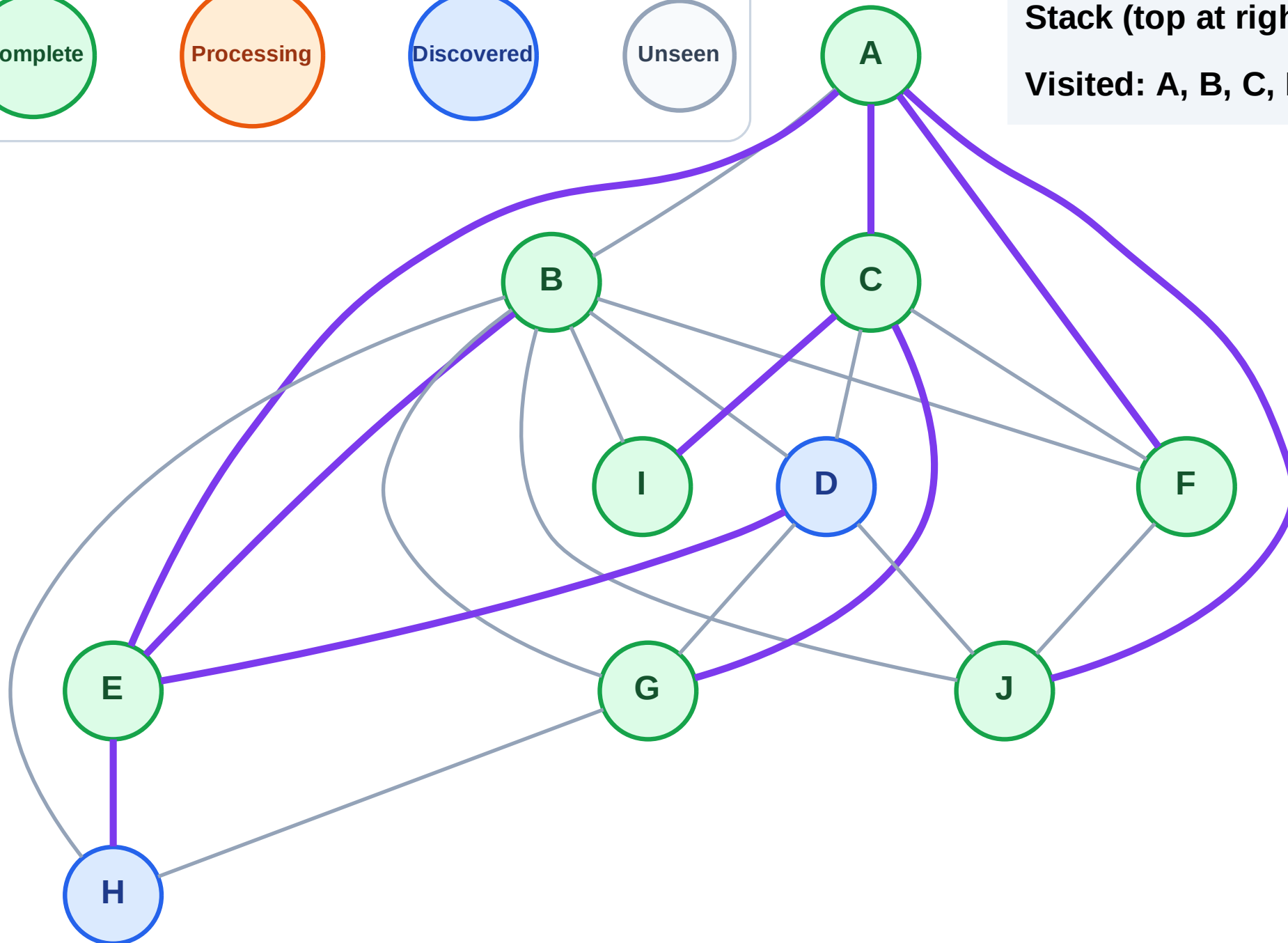
Step 17: No new neighbors from B

Legend



Stack (top at right): H | D

Visited: A, B, C, E, F, G, I, J



DFS Traversal

Step 18: Process node D

Legend

Complete

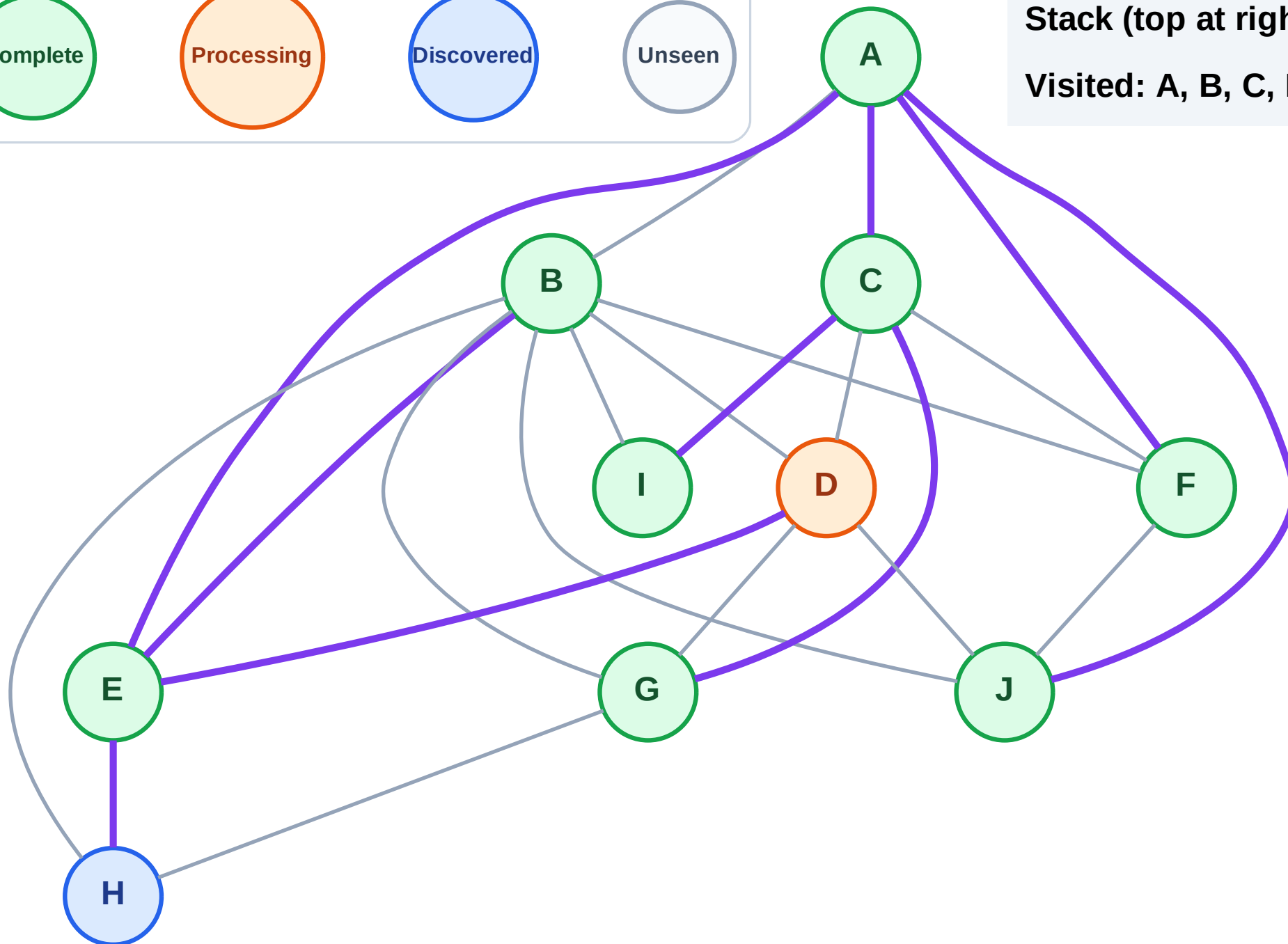
Processing

Discovered

Unseen

Stack (top at right): H

Visited: A, B, C, E, F, G, I, J



Step 19: No new neighbors from D

Legend

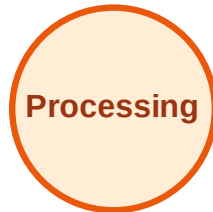
Unseen

Visited: A, B, C, D, E, F, G, I, J

DFS Traversal

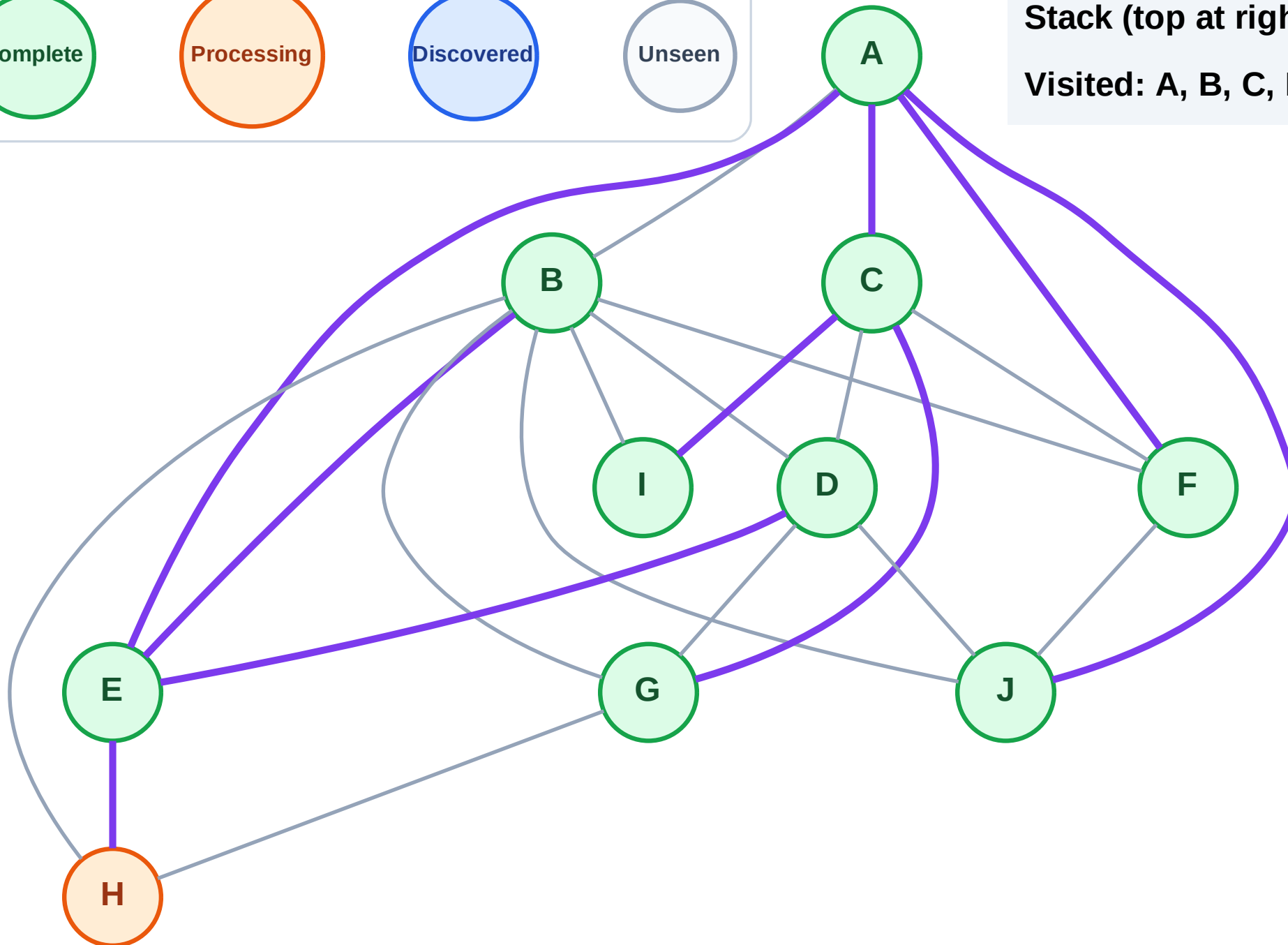
Step 20: Process node H

Legend



Stack (top at right): (empty)

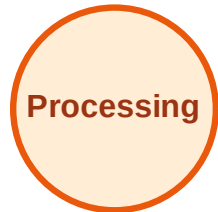
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

